Mechanics and Balance of Magic Gems: Radial Vector Laws, Symmetric-Top Precession, Nested Core Inclusions, and Mirror-Twin Towers

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Abstract

Lifting an $n \times n$ arrangement S of $\{1, \dots, n^2\}$ over the centred grid $A_n \subset \mathbb{R}^2$ produces n^2 three-dimensional vectors $\mathbf{p}_{ij}(\lambda) = (x_j, y_i, \lambda z_{ij})$ radiating from the origin O and their convex hull, the *Magic Gem* $\mathcal{G}_\lambda(S)$. We investigate when the combinatorial balance of a magic square translates into physical equilibrium, smooth rotation, and structural stability across four sculptural realizations: radial spoke stars and balance wheels from O, pivot-supported spinning tops, solid/skeletal/papercraft polyhedra, and freestanding stacked towers. First, by *length–weight duality*, scaling vector lengths along unit rays $\hat{\mathbf{p}}_{ij}(\lambda)$ by $L_{ij} = R_{ij}(\lambda)^\beta$ is mechanically equivalent to weighting $\mathbf{p}_{ij}(\lambda)$ by $w_{ij} = R_{ij}(\lambda)^{\beta-1}$. For cubic 3D lengths ($\beta = 3$), the vertical resultant $(\mathbf{V}_3)_z$ vanishes identically for every semi-magic square and every $\lambda > 0$, while on the unit sphere S^2 ($\beta = 0$), all 880 order-four magic squares admit positive minimum-variance balancing weights $\mathbf{w}^*(S) > \mathbf{0}$. In the plane, decomposing A_4 into three rings of radii $(\sqrt{2}, \sqrt{10}, 3\sqrt{2})/2$ proves that planar unit vectors ($\alpha = 0$) never balance on $\mathcal{M}_4$ because $r_2/r_1 = \sqrt{5} \notin \mathbb{Q}$; the global minimum $18\sqrt{10}/5 - 8\sqrt{2} \approx 0.07049$ is governed by the Pell convergent $9/4$ on eight Group VI squares, while 204 squares with parallel ring dipoles ($\kappa > 0$) admit a second balance exponent $\alpha^*(\kappa) \neq 1$ (25 at $\alpha^* = -1$ and 25 at $\alpha^* = +3$). Second, equal point masses make every semi-magic square an axisymmetric rotor with spherical-top scale $\lambda_{\text{iso}} = (n^2+1)^{-1/2}$ that spins torque-free on a central pivot under gravity, whereas entry masses $m_{ij} = s_{ij}$ split the horizontal moments by $2|Q_{xy}(S)|$ and trigger Dzhanibekov flips on 824 squares. Third, we classify the 51 naturally balanced solids ($\mathbf{c}_0 = \mathbf{0}$) and restore solid and skeletal balance via "diamond-in-a-diamond" convex layers $\mathcal{G}_0(S) \supset \mathcal{G}_1(S)$ (35 two-density solids, led by 8 printable with modest excess density $\Delta\rho_1/\rho_0 \in [5/4, 40/19]$) and nested ball-and-rod wireframes $G(\mathcal{G}_0) \cup G(\mathcal{G}_1)$ (158 gems at $\lambda = 1$). Moreover, we prove that the complete $\binom{N}{2}$-rod network K_N satisfies $\mathbf{c}(K_N) = \mathbf{0}$ and $\Sigma_{\text{rods}} = \frac{2N-3}{3(N-1)}\Sigma(\lambda)$ ($\frac{29}{45}\Sigma(\lambda)$ at $n = 4$), and show that every spanning-tree papercraft cutout net of $\partial\mathcal{G}_\lambda(S)$ has $N_{\text{tabs}} = f_0 - 1$ glue tabs and unfolds without overlap across all 880 gems. Finally, central inversion $\mathcal{G}_\lambda(r^2\overline{S}) = -\mathcal{G}_\lambda(S)$ locks any gem with centrally symmetric upper and lower facets onto its mirror twin to form zero-rotation vertical towers, led by 17 knight's-move gems (such as #763).

1 Introduction

Over five centuries ago, Albrecht Dürer contributed two enduring landmarks at the intersection of mathematics and the visual arts: the order-four magic square in his 1514 engraving *Melencolia I* (square #175 in Frénicle's 1693 census of the 880 order-four magic squares [8, 6, 1]) and the first printed polyhedral net unfoldings in his 1525 treatise *Underweysung der Messung* [7, 18, 5, 9]. Recall that an arrangement $S = (s_{ij})_{0 \le i,j < n}$ of the integers $\{1, \dots, N\}$, $N = n^2$, is a *semi-magic square* if every row and column sums to the magic constant $\mu_n = n(N+1)/2$, and a *magic square* if the two main diagonals also sum to μ_n. Measuring coordinates from the board centre and the mean entry $\bar{s} = (N+1)/2$,

$$x_j = j - \frac{n-1}{2}, \qquad y_i = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \frac{N+1}{2}, \tag{1}$$

turns each cell (i, j) into a three-dimensional point and radial vector from the origin $O = (0, 0, 0)$:

$$\mathbf{p}_{ij}(\lambda) = (x_j,\, y_i,\, \lambda z_{ij}) \in \mathbb{R}^3, \qquad P_S(\lambda) = \big\{\mathbf{p}_{ij}(\lambda) : 0 \le i, j < n\big\},$$

where $\lambda > 0$ is the vertical lift scale (Figure 2(a)). The convex hull

$$\mathcal{G}_\lambda(S) := \operatorname{conv} P_S(\lambda) \subset \mathbb{R}^3 \qquad \big(\mathcal{G}(S) := \mathcal{G}_1(S)\big)$$

is a three-dimensional convex polyhedron that we call the *Magic Gem* of S, reuniting Dürer's magic squares with his polyhedral geometry.

In classical work on the planar mechanics of magic squares, Loly [11] and Rogers and Loly [16, 17] interpreted the entries s_{ij} as point masses on the flat grid $\mathbf{r}_{ij} = (x_j, y_i)$, proving that the planar centre of mass lies at the origin O and that the polar moment of inertia $\sum_{i,j} s_{ij}(x_j^2 + y_i^2) = n^2(n^4-1)/12$ is invariant across all semi-magic squares of order n. Inspired by fabricating physical 3D-printed models, radial balance wheels, ball-and-rod wireframes, interconnected strut sculptures, and folded cardstock cutouts in the open-source `magicGemWeb` studio [12], this paper investigates one unifying physical and sculptural question across dimensions:

> *When a magic square is lifted into $\mathbb{R}^3$ and fabricated as a physical sculpture—whether as a 16-spoke radial star or weighted balance wheel from the origin (Section 2), a kinetic spinning top on a central pivot (Section 3), a cast solid, nested two-material core, 3D-printed wireframe, complete K_{16} strut network, or folded papercraft shell (Section 4), or a freestanding column of stacked blocks (Section 5)—when does the combinatorial balance of the magic square translate into physical equilibrium, smooth rotation, and structural stability?*

Figure 1 introduces the six flagship order-four magic squares (#175, #70, #45↔#54, #46, #673, and #763↔#756) that serve as our recurring physical exemplars across four progressive mechanical regimes:

1. **Radial spoke sculptures and balance wheels from the origin (Section 2).** Scaling 3D spoke lengths by $L_{ij} = R_{ij}(\lambda)^\beta$ is mechanically dual to weighting natural vectors $\mathbf{p}_{ij}(\lambda)$ by $w_{ij} = R_{ij}(\lambda)^{\beta-1}$ (Proposition 2.2). In 3D (Theorem 2.5, Figure 2), cubic spokes ($\beta = 3$) have a strictly vanishing vertical resultant $(\mathbf{V}_3)_z = 0$ for every semi-magic square, while on the unit sphere S^2 ($\beta = 0$) every order-four square admits strictly positive minimum-variance weights $\mathbf{w}^*(S) > \mathbf{0}$ restoring $\sum w^*_{ij}\hat{\mathbf{p}}_{ij} = \mathbf{0}$. In the grid plane (Theorem 2.3, Figure 3), decomposing A_4 into three concentric rings of radii $r_1 = \sqrt{2}/2$, $r_2 = \sqrt{10}/2$, and $r_3 = 3\sqrt{2}/2$ proves that unit-length planar spokes ($\alpha = 0$) can never balance because $r_2/r_1 = \sqrt{5} \notin \mathbb{Q}$; instead, $\min_{\mathcal{M}_4} \|\mathbf{B}_{\text{unit}}(S)\|_2 = \frac{18\sqrt{10}}{5} - 8\sqrt{2} \approx 0.0705$ is governed by the Pell convergent $9/4 \approx \sqrt{5}$ on eight Group VI squares led by #70 (Figure 1(b)), while 204 squares with parallel dipoles $\mathbf{d}_{\text{out}} = \kappa\mathbf{d}_{\text{in}}$ admit a second power-law root $\alpha^*(\kappa) \neq 1$, including 25 at $\alpha^* = -1$ (#45, Figure 1(c)) and 25 at $\alpha^* = +3$ (#286; Corollary 2.4).

2. **Three-dimensional rotation: symmetric-top precession and Dzhanibekov flips (Section 3).** For equal point masses at $\mathbf{p}_{ij}(\lambda)$, vanishing of the first moments makes the inertia tensor about O strictly axisymmetric with universal ratio $I_\perp(\lambda)/I_{zz} = \frac{1}{2}(1+\lambda^2(n^2+1))$ (Theorem 3.1, Figure 4(a)): every point-weighted gem passes through an exact *spherical-top transition* at $\lambda_{\text{iso}} = (n^2+1)^{-1/2}$ ($1/\sqrt{17} \approx 0.2425$ at $n = 4$) and a $9:1$ closed-orbit prolate regime at $\lambda = 1$. Conversely, entry-weighted masses $m_{ij} = s_{ij}$ (Theorem 3.2) split the horizontal principal moments by $2|Q_{xy}(S)| = 2|\sum z_{ij}x_jy_i|$ along $y = \pm x$ and couple the vertical products of inertia to $\mathbf{V}_3(S;\lambda)$, triggering 180° Dzhanibekov flips on 824 of the 880 squares (led by #46 at $|Q_{xy}| = 60$, Figure 1(d) and Figure 4(b–d)).

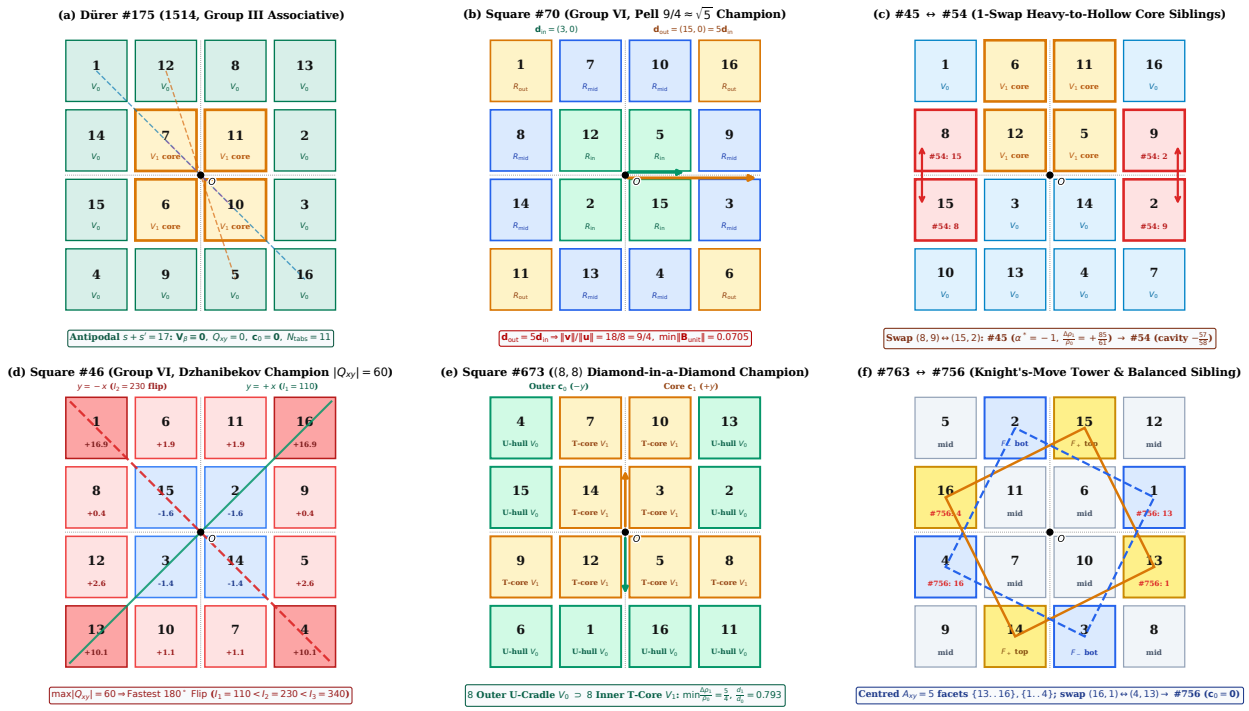


Figure 1: Visual anatomy of the six flagship order-four magic squares across Sections 2–5. (a) Dürer's associative square #175 (1514, Group III), whose antipodal pairs sum to 17 across O, partitioning into 12 outer hull vertices V_0 and 4 inner core vertices $V_1 = \{6, 7, 10, 11\}$. (b) Square #70 (Group VI), coloured by concentric rings $R_{\text{in}}, R_{\text{mid}}, R_{\text{out}}$, where parallel ring dipoles $\mathbf{d}_{\text{out}} = (15, 0) = 5\mathbf{d}_{\text{in}}$ achieve the fundamental Pell ratio $\|\mathbf{v}\|/\|\mathbf{u}\| = 18/8 = 9/4 \approx \sqrt{5}$. (c) Square #45 ($\kappa = +9$, inverse-radial root $\alpha^* = -1$, heavy core $\Delta\rho_1/\rho_0 = +85/61$), where swapping the single outer border pair $(8, 9) \leftrightarrow (15, 2)$ between rows 2 and 3 yields sibling #54 with the same $(V_0, V_1) = (10, 6)$ vertex partition but a nearly hollow core cavity ($\Delta\rho_1/\rho_0 = -57/58$). (d) Square #46 (Group VI), shaded by cell quadrupole contribution $z_{ij}x_jy_i$, attaining the global maximum $|Q_{xy}| = 60$ ($I_1 = 110 < I_2 = 230 < I_3 = 340$) and fastest 180° Dzhanibekov flip about $y = -x$. (e) Square #673 (Group VI), whose 8 outer vertices form a U-cradle V_0 enclosing an 8-vertex inverted-T core V_1, achieving the minimum solid excess density $\Delta\rho_1/\rho_0 = 5/4$. (f) Square #763 (Group VI), whose centred 2×1 knight's-move facets $\{13, 14, 15, 16\}$ and $\{1, 2, 3, 4\}$ lock into an area-5 mirror-twin tower, and whose single border-pair swap $(16, 1) \leftrightarrow (4, 13)$ produces the naturally balanced solid #756 ($\mathbf{c}_0 = \mathbf{0}$).

3. **Restoring solid, skeletal, and thin-shell balance via "diamond-in-a-diamond" layers, complete K_N networks, and papercraft nets (Section 4).** Casting a uniform solid $\mathcal{G}_0(S) = \text{conv}\, P_S$ shifts $\mathbf{c}_0(S)$ away from O on 829 of the 880 squares, leaving 51 naturally balanced solids (48 associative plus Group VI squares #86, #739, and #756; Proposition 4.1). Peeling $\mathcal{G}_0(S) \supset \mathcal{G}_1(S)$ (Proposition 4.2) yields 35 two-density balanced solids (Theorem 4.3, Figure 5): 33 with a denser inner core (led by #673 at $\Delta\rho_1/\rho_0 = 5/4$, Figure 1(e)) and 2 (#123 and #54, Figure 1(c)) where volumetric simplex weighting pulls $\mathbf{c}_0$ to the same side of O as $\mathbf{c}_1$, requiring a lighter or nearly hollow core cavity. For skeletal and sheet fabrications (Section 4.4, Figure 6), the complete $\binom{N}{2}$-rod network K_N with equal-mass rods has $\mathbf{c}(K_N) = \mathbf{0}$ and $\Sigma_{\text{rods}} = \frac{2N-3}{3(N-1)}\Sigma(\lambda) = \frac{29}{45}\Sigma(\lambda)$ at $n = 4$ (Theorem 4.4), tuning inner-to-outer rod diameters d_1/d_0 balances 158 Group VI wireframes at $\lambda = 1$ (164 at $\lambda = 0.20$), and every gem unfolds along a breadth-first dual spanning tree into a non-overlapping 2D papercraft net with $N_{\text{tabs}} = f_0 - 1$ glue tabs (Proposition 4.5).

4. **Mirror-twin facet locking and chiral vertical towers (Section 5).** While only 124 of the 880 gems have parallel top and bottom facets on the *same* gem, the half-turned complement $S^* = r^2\overline{S}$ satisfies $\mathcal{G}_\lambda(S^*) = -\mathcal{G}_\lambda(S)$ (Theorem 5.3), pairing every upper facet $F_+ \subset \mathcal{G}_\lambda(S)$ with a congruent parallel lower facet $-F_+ \subset \mathcal{G}_\lambda(S^*)$. Alternating unrotated

translates of $\mathcal{G}_\lambda(S)$ and $\mathcal{G}_\lambda(S^*)$ along centrally symmetric vertical-axis facets forms a zero-drift vertical tower for 48 gems (Proposition 5.4, Figure 7), led by 17 gems with area-5 centred knight's-move square facets such as chiral Group VI gem #763 (Figure 1(f), Remark 5.5).

Every theorem and census table in this paper is certified in exact integer and rational arithmetic (`fractions.Fraction`) by `anc/verify_paper_c.py` (Section A), and all five mechanical regimes can be explored in real time in the companion interactive browser visual essay (`web/index.html`).

Conventions, indexing, and relation to companion work. Let $A_n = \{(x_j, y_i) : 0 \le i, j < n\}$ denote the centred grid (1), let $a_n = (n^2-1)/12$ and $b_n = (n^4-1)/12 = (n^2+1)a_n$ denote the marginal variances of x_j and z_{ij}, and let the dihedral group $D_4 = \langle r, h\rangle$ (with 90° rotation r and horizontal/vertical reflections h, v) act on arrangements together with the complement involution $\tau(S) = \overline{S} = (N+1)\mathbf{1} - S$. An arrangement S is *associative* (Dudeney's Group III of 48 squares at $n = 4$, including Dürer's #175) if $r^2 S = \overline{S}$, i.e. $z(-x,-y) = -z(x,y)$, and belongs to Dudeney's *Group VI* (304 squares at $n = 4$) if $\overline{S}$ equals hS or vS, inducing a 180° half-turn symmetry ($C_2 \subset SO(3)$) of $\mathcal{G}_\lambda(S)$ in $\mathbb{R}^3$. Companion preprints [13, 14] catalogue the static polyhedral combinatorics and bivariate Prouhet–Tarry–Escott moment hierarchy of semi-magic squares; the present paper is self-contained. Order-four magic squares are numbered #1, …, #880 by their 1-based position in Frénicle's standard list (`anc/squares880.txt`); Tables 2 and 3 also list the 0-based index (#0 = #1 − 1).

2 Vectors from the Origin: Length–Weight Duality and the $\sqrt{5}$ Pell Barrier

In the fabrication suite `magicGemWeb` [12], the 16 cells of an order-four magic square are built as two radial sculptures pivoted at the origin O: (i) a *3D 16-spoke hub-and-strut star* (`magicGem.scad`, Figure 2(a)) with spokes $\mathbf{p}_{ij}(\lambda) = (x_j, y_i, \lambda z_{ij})$ of length $R_{ij}(\lambda) = \|\mathbf{p}_{ij}(\lambda)\|_2$ along unit directions $\hat{\mathbf{p}}_{ij}(\lambda) = \mathbf{p}_{ij}(\lambda)/R_{ij}(\lambda) \in S^2$; and (ii) a *planar 16-spoke balance wheel* on a conical fulcrum (`weighted_ring.stl`, Figure 3(a)) with spokes along $\mathbf{r}_{ij} = (x_j, y_i)$ of radius $r_{ij} = \|\mathbf{r}_{ij}\|_2$ and unit direction $\mathbf{u}_{ij} = \mathbf{r}_{ij}/r_{ij} \in S^1$. At linear spoke lengths ($L_{ij} = R_{ij}(\lambda)$ or $\ell(r_{ij}) = r_{ij}$), both sculptures balance identically at O because $\sum_{i,j} \mathbf{p}_{ij}(\lambda) = \mathbf{0}$ and $\sum_{i,j} z_{ij}\mathbf{r}_{ij} = \mathbf{0}$ [11, 17], closing the 16 planar vectors into a head-to-tail polygon (Figure 3(b)). Trimming the spokes to unit length ($\beta = 0$ on S^2 or $\alpha = 0$ on S^1), scaling lengths by power laws ($R_{ij}(\lambda)^\beta$ or r_{ij}^α), or varying the tip weights (w_{ij} or m_{ij}) tests when static equilibrium at O survives nonlinear radial distortion.

2.1 Scaling Vector Lengths from the Origin Is Dual to Reweighting Masses

Definition 2.1 (Generalized 3D and planar origin-vector resultants)**.** For an arrangement S of order n and vertical lift scale $\lambda > 0$, let $R_{ij}(\lambda) = \sqrt{x_j^2 + y_i^2 + \lambda^2 z_{ij}^2}$ and $\hat{\mathbf{p}}_{ij}(\lambda) = \mathbf{p}_{ij}(\lambda)/R_{ij}(\lambda)$ ($\mathbf{0}$ if $R_{ij}(\lambda) = 0$). Given cell weights $w = (w_{ij})$ and prescribed 3D vector lengths $L = (L_{ij})$, the *3D origin-vector resultant* is

$$\mathbf{V}_{w,L}(S;\lambda) = \sum_{i,j} w_{ij}\, L_{ij}\, \hat{\mathbf{p}}_{ij}(\lambda) = \sum_{R_{ij}(\lambda)>0} \left(\frac{w_{ij} L_{ij}}{R_{ij}(\lambda)}\right) \mathbf{p}_{ij}(\lambda) \in \mathbb{R}^3. \tag{2}$$

For equal weights $w_{ij} = 1$ and power-law 3D lengths $L_{ij} = R_{ij}(\lambda)^\beta$ ($\beta \in \mathbb{R}$), we write

$$\mathbf{V}_\beta(S;\lambda) := \sum_{i,j} R_{ij}(\lambda)^\beta\, \hat{\mathbf{p}}_{ij}(\lambda) = \sum_{i,j} R_{ij}(\lambda)^{\beta-1}\, \mathbf{p}_{ij}(\lambda) \in \mathbb{R}^3, \tag{3}$$

and for weighted unit vectors ($L_{ij} = 1$) on S^2, we write $\mathbf{B}^{(3)}_{w,\lambda}(S) := \mathbf{V}_{w,1}(S;\lambda) = \sum_{i,j} w_{ij}\hat{\mathbf{p}}_{ij}(\lambda)$. Similarly, in the grid plane, given cell masses $m = (m_{ij})$ and a radial length profile $\ell\colon [0,\infty) \to$

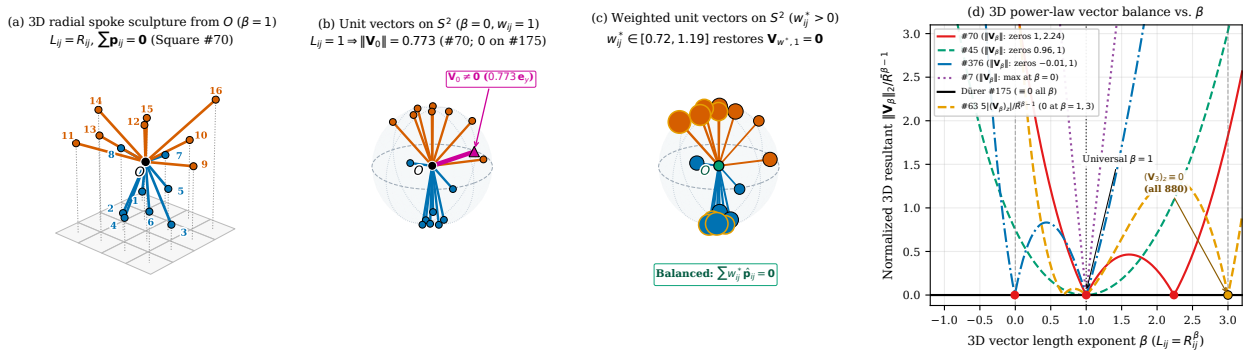


Figure 2: Three-dimensional radial spoke sculptures from the origin $O = (0, 0, 0)$ of variable length and tip weight. (a) The 16-spoke radial sculpture $\mathbf{p}_{ij}$ ($\beta = 1$, spoke length $L_{ij} = R_{ij}$, orange for $z_{ij} > 0$, blue for $z_{ij} < 0$) of non-associative square #70 over the floor grid, balancing identically at the central hub O ($\sum \mathbf{p}_{ij} = \mathbf{0}$). (b) Trimming all 16 spokes to unit length $L_{ij} = 1$ ($\beta = 0$, equal tip weights $w_{ij} = 1$) on the unit sphere S^2 leaves a non-zero resultant $\mathbf{V}_0 = (0, 0.773, 0)$ (magenta arrow) on square #70, whereas antipodal pairs cancel on associative Dürer #175. (c) Assigning the minimum-variance positive tip weights $w_{ij}^* \in [0.72, 1.19]$ (sphere tip sizes, gold-rimmed for $w_{ij}^* > 1$) restores exact unit-sphere balance $\sum w_{ij}^* \hat{\mathbf{p}}_{ij} = \mathbf{0}$ at O. (d) Normalized 3D power-law resultant $\|\mathbf{V}_\beta(S; 1)\|_2/\bar{R}^{\beta-1}$ versus length exponent β, showing universal balance at $\beta = 1$, second power-law roots $\beta^* \neq 1$ on Group VI squares (#70 at $\beta^* = 2.24$, #45 at 0.96, #376 at -0.008), and identical vertical vanishing $(\mathbf{V}_3)_z = 0$ at $\beta = 3$ (gold dashed curve showing $5|(\mathbf{V}_\beta)_z|/\bar{R}^{\beta-1}$ on square #63).

$[0, \infty)$ along unit rays $\mathbf{u}_{ij} = \mathbf{r}_{ij}/r_{ij}$, the *planar radial resultant* is

$$\mathbf{B}_{m,\ell}(S) = \sum_{i,j} m_{ij}\, \ell(r_{ij})\, \mathbf{u}_{ij} = \sum_{r_{ij}>0} \left(\frac{m_{ij}\ell(r_{ij})}{r_{ij}} \right) (x_j, y_i) \in \mathbb{R}^2. \tag{4}$$

Because $\ell(r_{ij})$ depends only on the radius r_{ij}, grid symmetry gives $\sum_{i,j} \ell(r_{ij})\mathbf{u}_{ij} = \mathbf{0}$, so replacing the positive entry masses $m_{ij} = s_{ij} = \bar{s} + z_{ij}$ by the signed heights $m_{ij} = z_{ij}$ leaves the planar resultant unchanged: $\mathbf{B}_{s,\ell}(S) = \mathbf{B}_{z,\ell}(S)$.

Proposition 2.2 (Length–weight duality and order-three radial law)**.** *Let $n \geq 3$ and $\lambda > 0$.*

1. (3D and 2D length–weight duality.) *Scaling 3D vector lengths from the origin to L_{ij} with weights w_{ij} produces the same resultant (2) as keeping the natural 3D vectors $\mathbf{p}_{ij}(\lambda)$ ($L'_{ij} = R_{ij}(\lambda)$) and assigning effective weights $w'_{ij} = w_{ij}L_{ij}/R_{ij}(\lambda)$. In particular:*

   (a) *Power-law 3D vector lengths $L_{ij} = R_{ij}(\lambda)^\beta$ with unit weights ($w_{ij} = 1$) are mechanically dual to natural 3D vectors $\mathbf{p}_{ij}(\lambda)$ with weights $w'_{ij} = R_{ij}(\lambda)^{\beta-1}$.*

   (b) *Unit-length 3D vectors on S^2 ($L_{ij} = 1$) balance identically ($\sum_{i,j} w_{ij}\hat{\mathbf{p}}_{ij}(\lambda) = \mathbf{0}$) for every arrangement S if weighted by their 3D lengths $w_{ij} = c\, R_{ij}(\lambda)$ ($c \in \mathbb{R}$).*

   (c) *In the grid plane, unit vectors $\mathbf{u}_{ij}$ balance for every semi-magic square ($\sum_{i,j} m_{ij}\mathbf{u}_{ij} = \mathbf{0}$) whenever $m_{ij} = c\, z_{ij} r_{ij}$, and a radial profile ℓ_{ij} with $m_{ij} = z_{ij}$ balances every semi-magic square if and only if $(\ell_{ij}/r_{ij})\mathbf{r}_{ij}$ has components in the row–column subspace $W_{\mathrm{rc}} = \{a(x_j) + b(y_i)\}$.*

2. (Order-three exact formula.) *At order $n = 3$, the eight non-central cells of A_3 lie on two concentric rings: four edge midpoints $R_1 = \{(\pm 1, 0), (0, \pm 1)\}$ ($r_1 = 1$) and four corners $R_{\sqrt{2}} = \{(\pm 1, \pm 1)\}$ ($r_2 = \sqrt{2}$). Every semi-magic square S of order 3 satisfies $\mathbf{d}_1(S) + \mathbf{d}_2(S) = \mathbf{0}$, where $\mathbf{d}_k(S) = \sum_{(i,j)\in R_k} z_{ij}\mathbf{r}_{ij}$, and hence*

$$\mathbf{B}_{z,\ell}(S) = \left(\ell(1) - \frac{\ell(\sqrt{2})}{\sqrt{2}} \right) \mathbf{d}_1(S). \tag{5}$$

For all eight magic squares of order 3 (the D_4-orbit of the Lo Shu square), $\|\mathbf{d}_1(S)\|_2 = \sqrt{80} = 4\sqrt{5}$, giving

$$\|\mathbf{B}_{z,\ell}(S)\|_2 = 4\sqrt{5}\left|\ell(1) - \frac{\ell(\sqrt{2})}{\sqrt{2}}\right|, \qquad \|\mathbf{B}_{\text{unit}}(S)\|_2 = 4\sqrt{5} - 2\sqrt{10} \approx 2.619717. \tag{6}$$

Thus $\mathbf{B}_{z,\ell}(S) = \mathbf{0}$ at $n = 3$ if and only if $\ell(\sqrt{2}) = \sqrt{2}\,\ell(1)$.

Proof. Part (1) follows directly from (2)–(4), the identity $\sum_{i,j} \mathbf{p}_{ij}(\lambda) = \mathbf{0}$, and the characterization of W_{rc} as the orthogonal complement of the zero-line-sum space [14, Sect. 2]. In Part (2), $\mathbf{r}_{11} = (0,0)$ implies $\mathbf{d}_1(S) + \mathbf{d}_2(S) = \sum_{i,j} z_{ij}\mathbf{r}_{ij} = \mathbf{0}$ for every semi-magic square, which yields (5) upon grouping (4) over R_1 and $R_{\sqrt{2}}$. Evaluating $\mathbf{d}_1 = (-8, 4)$ on the Lo Shu square gives $\|\mathbf{d}_1\|_2 = \sqrt{80} = 4\sqrt{5}$, and D_4-invariance of the Euclidean norm yields (6). □

2.2 Unit Vectors on the 4×4 Grid Cannot Balance, and Eight Group VI Squares Attain the 9/4 Pell Minimum

At order $n = 4$, the sixteen cells of $A_4 = \{-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\}^2$ partition into three concentric rings around the origin O (Figure 1(b), Figure 3(a)):

$$\begin{aligned} R_{\text{in}} &= \{(\pm\tfrac{1}{2}, \pm\tfrac{1}{2})\} && \text{(4 inner cells,} \quad r_1 = \tfrac{\sqrt{2}}{2}\text{),} \\ R_{\text{mid}} &= \{(\pm\tfrac{3}{2}, \pm\tfrac{1}{2}),\ (\pm\tfrac{1}{2}, \pm\tfrac{3}{2})\} && \text{(8 edge-interior cells,} \quad r_2 = \tfrac{\sqrt{10}}{2}\text{),} \\ R_{\text{out}} &= \{(\pm\tfrac{3}{2}, \pm\tfrac{3}{2})\} && \text{(4 corner cells,} \quad r_3 = \tfrac{3\sqrt{2}}{2}\text{).} \end{aligned}$$

Whereas the outer and inner radii have rational ratio $r_3/r_1 = 3$, the middle and inner radii have irrational ratio $r_2/r_1 = \sqrt{5} \notin \mathbb{Q}$. This irrationality governs the planar vector mechanics of all 880 order-four magic squares.

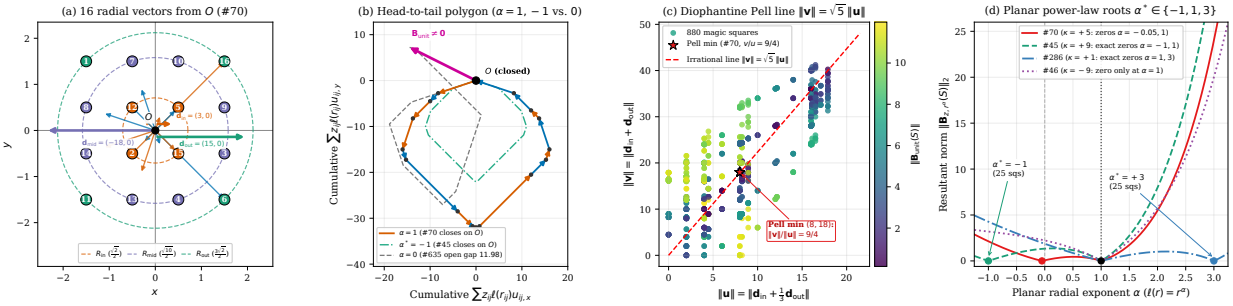


Figure 3: Planar radial vectors from the origin $O = (0,0)$, head-to-tail polygon closure, and the Diophantine Pell minimum at order four. (a) The 16 radial vectors $z_{ij}\mathbf{r}_{ij}$ radiating from O (orange $z_{ij} > 0$ outward, blue $z_{ij} < 0$ inward) across the three concentric rings $R_{\text{in}}, R_{\text{mid}}, R_{\text{out}}$ on square #70, together with the three ring dipole resultants $\mathbf{d}_{\text{in}} = (3,0)$, $\mathbf{d}_{\text{out}} = (15,0)$, and $\mathbf{d}_{\text{mid}} = (-18,0)$. (b) Placing the 16 vectors $z_{ij}\ell(r_{ij})\mathbf{u}_{ij}$ head-to-tail in order of angle around O produces a closed 16-gon at linear lengths $\alpha = 1$ ($\ell(r) = r$ on #70, solid orange/blue) and at inverse radial lengths $\alpha^* = -1$ ($\ell(r) = 1/r$ on #45, green dash-dot), whereas unit lengths $\alpha = 0$ ($\ell(r) = 1$ on #635, dashed) fail to close by the magenta gap vector $\mathbf{B}_{\text{unit}}(S)$. (c) Scatter plot of $\|\mathbf{v}(S)\|_2 = \|\mathbf{d}_{\text{in}} + \mathbf{d}_{\text{out}}\|_2$ versus $\|\mathbf{u}(S)\|_2 = \|\mathbf{d}_{\text{in}} + \frac{1}{3}\mathbf{d}_{\text{out}}\|_2$ across all 880 magic squares against the irrational balance line $\|\mathbf{v}\| = \sqrt{5}\,\|\mathbf{u}\|$, with the eight Pell-minimum squares ($v/u = 9/4$) marked by red stars. (d) Resultant norm $\|\mathbf{B}_{z,r^\alpha}(S)\|_2$ versus power-law exponent α, highlighting the universal root at $\alpha = 1$ and the exact second roots at $\alpha^* = -1$ (#45, $\kappa = +9$), $\alpha^* = -0.050$ (#70, $\kappa = +5$), and $\alpha^* = +3$ (#286, $\kappa = +1$).

Theorem 2.3 (Three-Ring Decomposition, Irrationality No-Go, and Pell Minimum)**.** *For any arrangement S of order 4, define the three ring dipole vectors radiating from O:*

$$\mathbf{d}_{\text{in}}(S) = \sum_{c \in R_{\text{in}}} z_c \mathbf{r}_c, \qquad \mathbf{d}_{\text{mid}}(S) = \sum_{c \in R_{\text{mid}}} z_c \mathbf{r}_c, \qquad \mathbf{d}_{\text{out}}(S) = \sum_{c \in R_{\text{out}}} z_c \mathbf{r}_c.$$

1. (Ring sums, integrality, and non-vanishing.) *For every* 4×4 *arrangement* S *of distinct integers,* $\mathbf{d}_{\text{in}}(S) \neq \mathbf{0}$ *and* $\mathbf{d}_{\text{out}}(S) \neq \mathbf{0}$. *For every order-four magic square* $S \in \mathcal{M}_4$, *both* R_{in} *and* R_{out} *sum to* $\mu_4 = 34$ *(so* $\sum_{c \in R_{\text{in}}} z_c = \sum_{c \in R_{\text{out}}} z_c = 0$*), and hence* $\mathbf{d}_{\text{in}}(S) \in \mathbb{Z}^2 \setminus \{\mathbf{0}\}$ *and* $\mathbf{d}_{\text{out}}(S) \in (3\mathbb{Z})^2 \setminus \{\mathbf{0}\}$, *with*

$$\|\mathbf{d}_{\text{in}}(S)\|_2 \geq 1, \qquad \|\mathbf{d}_{\text{out}}(S)\|_2 \geq 3. \tag{7}$$

2. (Three-ring formula.) *Whenever* $\sum_{c \in A_4} z_c \mathbf{r}_c = \mathbf{0}$ *(in particular, for every semi-magic square),* $\mathbf{d}_{\text{mid}}(S) = -\big(\mathbf{d}_{\text{in}}(S) + \mathbf{d}_{\text{out}}(S)\big)$, *and for every radial length profile* $\ell = (\ell_1, \ell_2, \ell_3)$ *on* (r_1, r_2, r_3),

$$\mathbf{B}_{z,\ell}(S) = \left(\frac{\ell_1}{r_1} - \frac{\ell_2}{r_2}\right) \mathbf{d}_{\text{in}}(S) + \left(\frac{\ell_3}{r_3} - \frac{\ell_2}{r_2}\right) \mathbf{d}_{\text{out}}(S). \tag{8}$$

3. (No-go theorem for unit vectors and even power laws.) *For planar unit vectors (*$\ell_1 = \ell_2 = \ell_3 = 1$, *i.e.* $\alpha = 0$*), substituting* $1/r_1 = \sqrt{2}$, $1/r_3 = \sqrt{2}/3$, *and* $1/r_2 = \sqrt{2}/\sqrt{5}$ *into (8) separates the rational* $\sqrt{2}$ *coefficients from the irrational* $\sqrt{2}/\sqrt{5}$ *coefficient:*

$$\mathbf{B}_{\text{unit}}(S) = \sqrt{2} \left(\mathbf{u}(S) - \frac{1}{\sqrt{5}} \mathbf{v}(S)\right), \tag{9}$$

where $\mathbf{u}(S) = \mathbf{d}_{\text{in}}(S) + \frac{1}{3}\mathbf{d}_{\text{out}}(S) \in \mathbb{Z}^2$ *collects the inner and outer ring contributions and* $\mathbf{v}(S) = \mathbf{d}_{\text{in}}(S) + \mathbf{d}_{\text{out}}(S) = -\mathbf{d}_{\text{mid}}(S) \in \mathbb{Z}^2$ *is the negative middle-ring dipole. Because* $\sqrt{5} \notin \mathbb{Q}$ *and* $\mathbf{d}_{\text{in}}(S), \mathbf{d}_{\text{out}}(S) \neq \mathbf{0}$, no 4×4 arrangement of distinct integers with vanishing first moments (and hence no 4×4 semi-magic or magic square) ever satisfies $\mathbf{B}_{\text{unit}}(S) = \mathbf{0}$, *nor can* $\mathbf{B}_{z,r^\alpha}(S) = \mathbf{0}$ *hold for any even integer* $\alpha \in 2\mathbb{Z}$.

4. (Exact Diophantine Pell minimum over $\mathcal{M}_4$.) *Across all* 880 *magic squares of order* 4, *the global minimum of* $\|\mathbf{B}_{\text{unit}}(S)\|_2$ *is*

$$\min_{S \in \mathcal{M}_4} \|\mathbf{B}_{\text{unit}}(S)\|_2 = \frac{18\sqrt{10}}{5} - 8\sqrt{2} = \frac{2\sqrt{10}}{5(9 + 4\sqrt{5})} = \frac{2\sqrt{10}}{5\varphi^6} \approx 0.07049108, \tag{10}$$

where $\varphi = (1+\sqrt{5})/2$. *This minimum is attained when* $\mathbf{d}_{\text{out}}(S) = 5\mathbf{d}_{\text{in}}(S)$ *with* $\|\mathbf{d}_{\text{in}}(S)\|_2 = 3$, *where* $\mathbf{u}(S)$ *and* $\mathbf{v}(S)$ *are parallel in the ratio* $\|\mathbf{v}\|/\|\mathbf{u}\| = 18/8 = 9/4$, *the fundamental Pell convergent to* $\sqrt{5}$ *(*$9^2 - 5 \cdot 4^2 = 1$*). Up to* D_4*-symmetry, exactly eight magic squares achieve (10): squares #70, #76, #121, #188, #452, #459, #490, and #574, all in Dudeney's Group VI.*

Proof. (1) Labelling the four entries of R_{in} at $(-\frac{1}{2}, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{2})$, $(-\frac{1}{2}, -\frac{1}{2})$, $(\frac{1}{2}, -\frac{1}{2})$ by $a, b, c, d \in \mathbb{Z}$ gives $\mathbf{d}_{\text{in}}(S) = \frac{1}{2}\big((b+d)-(a+c), (a+b)-(c+d)\big)$. If $\mathbf{d}_{\text{in}}(S) = \mathbf{0}$, then $b+d = a+c$ and $a+b = c+d$; adding and subtracting these equations forces $b = c$ and $a = d$, contradicting that S has distinct entries. For $S \in \mathcal{M}_4$, subtracting the sum of all 16 cells ($4\mu_4$) from the sum of the two outer rows and two outer columns ($R_0 + R_3 + C_0 + C_3 = 4\mu_4$) gives $\sum_{c \in R_{\text{out}}} s_c - \sum_{c \in R_{\text{in}}} s_c = 0$, while the two main diagonals sum to $D_1 + D_2 = \sum_{c \in R_{\text{out}}} s_c + \sum_{c \in R_{\text{in}}} s_c = 2\mu_4$; hence $\sum_{c \in R_{\text{in}}} s_c = \sum_{c \in R_{\text{out}}} s_c = \mu_4 = 34$ (an even integer), which implies $\mathbf{d}_{\text{in}}(S) \in \mathbb{Z}^2 \setminus \{\mathbf{0}\}$. Since the corner coordinates are $(\pm\frac{3}{2}, \pm\frac{3}{2}) = 3(\pm\frac{1}{2}, \pm\frac{1}{2})$, identical algebra gives $\mathbf{d}_{\text{out}}(S) \in (3\mathbb{Z})^2 \setminus \{\mathbf{0}\}$ and (7).

(2) Partitioning $A_4 = R_{\text{in}} \sqcup R_{\text{mid}} \sqcup R_{\text{out}}$ gives $\mathbf{d}_{\text{mid}} = -(\mathbf{d}_{\text{in}} + \mathbf{d}_{\text{out}})$, and substituting this relation into (4) yields (8).

(3) Setting $\ell_1 = \ell_2 = \ell_3 = 1$ with $1/r_1 = \sqrt{2}$, $1/r_2 = \sqrt{2}/\sqrt{5}$, and $1/r_3 = \sqrt{2}/3$ in (8) gives $\mathbf{B}_{\text{unit}}(S) = (\sqrt{2} - \sqrt{2}/\sqrt{5})\mathbf{d}_{\text{in}} + (\sqrt{2}/3 - \sqrt{2}/\sqrt{5})\mathbf{d}_{\text{out}} = \sqrt{2}\big((\mathbf{d}_{\text{in}} + \frac{1}{3}\mathbf{d}_{\text{out}}) - \frac{1}{\sqrt{5}}(\mathbf{d}_{\text{in}} + \mathbf{d}_{\text{out}})\big)$, which is (9). Because $\mathbf{u}(S), \mathbf{v}(S) \in \mathbb{Z}^2$ by (1) and $\sqrt{5} \notin \mathbb{Q}$, $\mathbf{B}_{\text{unit}}(S) = \mathbf{0}$ requires $\mathbf{u}(S) = \mathbf{v}(S) = \mathbf{0}$, which forces $\mathbf{d}_{\text{in}}(S) = \mathbf{d}_{\text{out}}(S) = \mathbf{0}$, contradicting (1). The same irrationality holds for any even exponent $\alpha = 2m$, where $r_2^{\alpha-1}/r_1^{\alpha-1} = 5^{m-1}\sqrt{5} \notin \mathbb{Q}$.

(4) Since $\mathbf{v}(S) - \mathbf{u}(S) = \frac{2}{3}\mathbf{d}_{\text{out}}(S) \in (2\mathbb{Z})^2$, the integer vectors satisfy the parity congruence $\mathbf{u}(S) \equiv \mathbf{v}(S) \pmod 2$ for every semi-magic square, while $\mathbf{d}_{\text{in}}, \mathbf{d}_{\text{out}} \neq \mathbf{0}$ implies $\mathbf{u} \neq \mathbf{v}$ and $\mathbf{v} \neq 3\mathbf{u}$.

Decomposing $\mathbf{v}$ parallel and orthogonal to $\mathbf{u}$ shows that $\|\mathbf{u} - \mathbf{v}/\sqrt{5}\|_2$ is minimized when $\mathbf{u} = p\mathbf{e}$ and $\mathbf{v} = q\mathbf{e}$ are collinear along a primitive lattice direction $\mathbf{e} = (a, b) \in \mathbb{Z}^2$ ($\gcd(a, b) = 1$), where $\|\mathbf{B}_{\text{unit}}(S)\|_2 = \sqrt{2(a^2+b^2)}\,|p - q/\sqrt{5}|$ and the prefactor $\sqrt{2(a^2+b^2)}$ is minimal along the coordinate axes ($a^2 + b^2 = 1$). Over all integer pairs with $p \equiv q \pmod 2$, the best approximations to $\sqrt{5}$ are governed by the Pell–Lucas equation $q^2 - 5p^2 = \pm 4$, whose positive solutions $(p, q) = (F_k, L_k)$ alternate between odd–odd Fibonacci–Lucas semiconvergents such as $(3, 7)$, $(5, 11)$, $(13, 29)$, $(21, 47)$ ($k \not\equiv 0 \pmod 3$) and even–even pairs $(F_{3m}, L_{3m}) = 2(p_m, q_m)$ obtained by doubling the principal continued-fraction convergents $q_m/p_m \in \{2/1, 9/4, 38/17, \dots\}$ of $\sqrt{5} = [2; \overline{4}]$. On $\mathcal{M}_4$, every pair (F_k, L_k) with $k \geq 7$ is excluded on two independent grounds:

(i) *Direct axial collinear bound:* Since $\frac{1}{3}\mathbf{d}_{\text{out}} = \frac{1}{2}(\mathbf{v} - \mathbf{u})$ and $\mathbf{d}_{\text{in}} = \frac{1}{2}(3\mathbf{u} - \mathbf{v})$, realizing $(p, q) = (F_k, L_k)$ along a coordinate axis requires $\frac{1}{3}|d_{\text{out}}| = \frac{1}{2}(L_k - F_k)$. Across $\mathcal{M}_4$, $\frac{1}{3}\|\mathbf{d}_{\text{out}}\|_\infty \leq 12$, which immediately excludes all $k \geq 8$ ($\frac{1}{2}(L_k - F_k) \geq \frac{1}{2}(47 - 21) = 13 > 12$). For $k = 7$, $(F_7, L_7) = (13, 29)$ requires $\frac{1}{3}|d_{\text{out}}| = \frac{1}{2}(29 - 13) = 8$ and $|d_{\text{in}}| = \frac{1}{2}(39 - 29) = 5$; however, saturating $\frac{1}{3}|d_{\text{out}}| = 8$ along a coordinate axis on $\mathcal{M}_4$ constrains the inner-ring dipole on every axial collinear square ($\mathbf{d}_{\text{in}} \parallel \mathbf{d}_{\text{out}}$) to $|d_{\text{in}}| \leq 4$ and hence $p = |d_{\text{in}} + \frac{1}{3}d_{\text{out}}| \leq 12 < 13$.

(ii) *Even-lattice property:* Subtracting $C_1 + C_2 = 2\mu_4$ and $C_0 + C_3 = 2\mu_4$ from $D_1 + D_2 = 2\mu_4$ gives $s_{00} + s_{03} = s_{31} + s_{32}$ and $s_{11} + s_{12} = s_{20} + s_{23}$, and across all 24 modulo-2 parity patterns of $\mathcal{M}_4$ the four top diagonal cells $T_{\text{diag}} = \{(0,0), (0,3), (1,1), (1,2)\}$ (and likewise $R_{\text{diag}} = \{(0,3), (3,3), (1,2), (2,2)\}$) always have an even sum $\sum_{c \in T_{\text{diag}}} s_c \in \{18, 20, \dots, 42\}$. Because $u_y(S) = \sum_{c \in T_{\text{diag}}} s_c - 34$ and $u_x(S) = \sum_{c \in R_{\text{diag}}} s_c - 34$, every magic square $S \in \mathcal{M}_4$ satisfies

$$\mathbf{u}(S) = 2\tilde{\mathbf{u}}(S) \in (2\mathbb{Z})^2, \quad \mathbf{v}(S) = 2\tilde{\mathbf{v}}(S) \in (2\mathbb{Z})^2, \quad \|\mathbf{u}(S)\|_\infty \leq 16, \quad \|\mathbf{u}(S)\|_2 \leq 8\sqrt{5} \approx 17.89,$$

excluding all odd–odd Pell–Lucas pairs and reducing (9) to $\|\mathbf{B}_{\text{unit}}(S)\|_2 = 2\sqrt{2}\,\|\tilde{\mathbf{u}}(S) - \tilde{\mathbf{v}}(S)/\sqrt{5}\|_2$ with $\tilde{\mathbf{u}}(S), \tilde{\mathbf{v}}(S) \in \mathbb{Z}^2$ and $\|\tilde{\mathbf{u}}(S)\|_2 \leq 4\sqrt{5} < 9$.

By Lagrange's theorem on continued fractions, across all integer denominators $1 \leq \tilde{u} < 17$, the unique best rational approximation to $\sqrt{5}$ is the fundamental Pell convergent $\tilde{v}/\tilde{u} = 9/4$ ($9^2 - 5{\cdot}4^2 = 1$), corresponding to $(p, q) = (F_6, L_6) = (8, 18)$ and giving $2\sqrt{2}(9/\sqrt{5} - 4) = 18\sqrt{10}/5 - 8\sqrt{2} = 2\sqrt{10}/(5\varphi^6) \approx 0.07049108$ (since $\varphi^6 = (2 + \sqrt{5})^2 = 9 + 4\sqrt{5}$), whereas the next convergent $38/17$ requires $\|\tilde{\mathbf{u}}\|_2 \geq 17 > 4\sqrt{5}$. On square #70 (Figure 1(b)), the inner ring $\{5, 12, 2, 15\}$ gives $\mathbf{d}_{\text{in}} = \frac{1}{2}\big((12 + 15) - (5 + 2), (12 + 5) - (2 + 15)\big) = (3, 0)$ and the outer corners $\{1, 16, 11, 6\}$ give $\mathbf{d}_{\text{out}} = \frac{3}{2}\big((16 + 6) - (1 + 11), (1 + 16) - (11 + 6)\big) = (15, 0) = 5\mathbf{d}_{\text{in}}$, realizing $\mathbf{u} = (8, 0)$, $\mathbf{v} = (18, 0)$, and (10). □

Equation (8) also characterizes when a non-linear power law $\ell(r) = r^\alpha$ ($\alpha \neq 1$) achieves exact planar balance (Figure 3(d)).

Corollary 2.4 (Non-linear planar power-law balance exponents $\alpha^* \neq 1$)**.** *For power-law radial lengths $\ell(r) = r^\alpha$ ($\alpha \in \mathbb{R}$), equation (8) takes the form*

$$\mathbf{B}_{z,r^\alpha}(S) = r_1^{\alpha-1}\Big[\big(1 - 5^{(\alpha-1)/2}\big)\mathbf{d}_{\text{in}}(S) + \big(3^{\alpha-1} - 5^{(\alpha-1)/2}\big)\mathbf{d}_{\text{out}}(S)\Big]. \tag{11}$$

1. *Because $1 < \sqrt{5} < 3$, the ratio $h(\alpha) = \frac{5^{(\alpha-1)/2} - 1}{3^{\alpha-1} - 5^{(\alpha-1)/2}}$ (extended by $h(1) = \frac{\ln 5}{\ln(9/5)} \approx 2.7381$) is a strictly decreasing bijection from $\mathbb{R}$ onto $(0, \infty)$. Consequently, an order-four magic square S possesses a second power-law balance exponent $\alpha^*(S) \neq 1$ satisfying $\mathbf{B}_{z,r^{\alpha^*}}(S) = \mathbf{0}$ if and only if $\mathbf{d}_{\text{in}}(S)$ and $\mathbf{d}_{\text{out}}(S)$ are strictly parallel ($\mathbf{d}_{\text{out}}(S) = \kappa\mathbf{d}_{\text{in}}(S)$ with $\kappa > 0$), which holds for exactly 204 of the 880 magic squares (while 204 squares have anti-parallel dipoles $\kappa < 0$, including 25 with $\kappa = -1$, i.e. $\mathbf{d}_{\text{mid}} = \mathbf{0}$).*

2. *Across all 880 magic squares, the second root $\alpha^*(\kappa)$ is an integer in only two cases:*

- $\alpha^* = -1$ *(inverse radial lengths* $\ell(r) = 1/r$*), corresponding to* $\kappa = h(-1) = \frac{5^{-1}-1}{3^{-2}-5^{-1}} = +9$*, achieved by exactly* 25 *squares (e.g. square #45 in Figure 1(c), where* $\mathbf{d}_{\text{in}} = (0, -2)$ *and* $\mathbf{d}_{\text{out}} = (0, -18) = 9\mathbf{d}_{\text{in}}$*);*
- $\alpha^* = +3$ *(cubic radial lengths* $\ell(r) = r^3$*), corresponding to* $\kappa = h(3) = \frac{5-1}{9-5} = +1$*, achieved by exactly* 25 *squares (e.g. square #286).*

For the 17 *squares with* $\kappa = +5$ *(including the eight Pell-minimum squares of Theorem 2.3(4)), the exact zero occurs at* $\alpha^*(5) \approx -0.049635$*, explaining why* $\|\mathbf{B}_{z,r^\alpha}(S)\|_2$ *pinches nearly to zero at* $\alpha = 0$ *(Figure 3(d)).*

Proof. Setting (11) to $\mathbf{0}$ for $\alpha \neq 1$ requires $\mathbf{d}_{\text{out}}(S) = h(\alpha)\mathbf{d}_{\text{in}}(S)$. Since $1 < 5^{t/2} < 3^t$ for $t = \alpha - 1 > 0$ and $3^t < 5^{t/2} < 1$ for $t < 0$, both the numerator and denominator of $h(\alpha)$ have the sign of $\alpha - 1$, so $h(\alpha) > 0$ with $\lim_{\alpha\to-\infty} h(\alpha) = +\infty$ and $\lim_{\alpha\to+\infty} h(\alpha) = 0$. Since $\kappa \in \mathbb{Q}_{>0}$ cannot equal the transcendental number $\frac{\ln 5}{\ln(9/5)}$, every square with $\kappa > 0$ has a unique root $\alpha^*(\kappa) = h^{-1}(\kappa) \neq 1$. By Theorem 2.3(3), an integer root $\alpha^* \neq 1$ must be an odd integer $\alpha^* = 2m + 1$ ($m \in \mathbb{Z} \setminus \{0\}$), where $h(2m + 1) = \frac{5^m-1}{9^m-5^m}$; comparing this sequence against the 15 positive rational ratios $\kappa \in \{\frac{1}{2}, \frac{3}{4}, 1, \frac{6}{5}, \frac{3}{2}, \frac{9}{5}, \frac{7}{3}, 3, \frac{27}{7}, 5, 6, \frac{15}{2}, 9, 12, 18\}$ realized on $\mathcal{M}_4$ leaves only $m = -1$ ($\kappa = 9$, 25 squares) and $m = +1$ ($\kappa = 1$, 25 squares). □

2.3 Three-Dimensional Vectors from the Origin Balance on the Unit Sphere Under Antipodal Symmetry or Positive Reweighting

Returning to three dimensions, we analyze the power-law resultant $\mathbf{V}_\beta(S; \lambda)$ in (3) and the unit-sphere resultant $\mathbf{B}^{(3)}_{w,\lambda}(S) = \sum_{i,j} w_{ij}\hat{\mathbf{p}}_{ij}(\lambda)$ (Figure 2(b–d)).

Theorem 2.5 (3D Power-Law Vectors, Cubic Vertical Vanishing, and Positive Unit-Sphere Weights)**.** *Let* S *be a semi-magic square of order* n *and let* $\lambda > 0$*.*

1. (Cubic 3D vector law $\beta = 3$.) *For cubic 3D vector lengths* $L_{ij} = R_{ij}(\lambda)^3$ *from the origin (mechanically equivalent by Proposition 2.2(1) to weighting the natural vectors* $\mathbf{p}_{ij}(\lambda)$ *by* $w_{ij} = R_{ij}(\lambda)^2 = r_{ij}^2 + \lambda^2 z_{ij}^2$*), the vertical* z*-component vanishes identically for every semi-magic square of any order* n *and every* $\lambda > 0$*:*

$$\mathbf{V}_3(S; \lambda) = \sum_{i,j} (r_{ij}^2 + \lambda^2 z_{ij}^2)(x_j, y_i, \lambda z_{ij}) = \left(\lambda^2 \sum_{i,j} z_{ij}^2 x_j,\ \ \lambda^2 \sum_{i,j} z_{ij}^2 y_i,\ \ 0\right), \tag{12}$$

whose horizontal vector $\sum_{i,j} z_{ij}^2(x_j, y_i)$ *is the quadratic height-dipole moment of [14] (and governs the vertical products of inertia of 3D entry-weighted gems in Theorem 3.2(3)). Because* λ^2 *factors out of* (12)*, at order* $n = 4$ *the cubic resultant* $\mathbf{V}_3(S; \lambda)$ *vanishes for some* $\lambda > 0$ *(and hence for all* $\lambda > 0$*) if and only if* S *is one of the* 48 *associative magic squares; across all* 880 *squares at* $\lambda = 1$*,* $\|\mathbf{V}_3(S; 1)\|_2 \in [0, 312]$ *with mean* 112.81*.*

2. (Unit-sphere parity and associative invariance at $\beta = 0$.) *Under signed height weights* $w_{ij} = z_{ij}$*, the vertical component* $(\mathbf{B}^{(3)}_{z,\lambda}(S))_z = \sum_{i,j} \lambda z_{ij}^2 / R_{ij}(\lambda) > 0$ *is strictly positive, so* $\mathbf{B}^{(3)}_{z,\lambda}(S)$ *never vanishes. Under any even weight function* $w_{ij} = f(|z_{ij}|)$ *(such as equal weights* $w_{ij} = 1$ *or* $w_{ij} = |z_{ij}|$*), every associative square satisfies* $\hat{\mathbf{p}}_{-c}(\lambda) = -\hat{\mathbf{p}}_c(\lambda)$ *and hence* $\mathbf{B}^{(3)}_{f(|z|),\lambda}(S) = \mathbf{0}$ *and* $\mathbf{V}_\beta(S; \lambda) = \mathbf{0}$ *for all* $\beta \in \mathbb{R}$ *and* $\lambda > 0$*. At* $n = 4$*,* $\mathbf{V}_0(S; \lambda) = \mathbf{0}$ *holds for all* $\lambda > 0$ *if and only if* S *is one of the* 48 *associative squares; across the remaining* 832 *non-associative squares at* $\lambda = 1$*,* $\|\mathbf{V}_0(S; 1)\|_2$ *ranges from* 0.00718 *(square #376) to* 2.4284 *(square #7) with mean* 0.9415*. Moreover, among the* 304 *self-complementary squares in Dudeney's Group VI (where* C_2 *symmetry confines* $\mathbf{V}_\beta(S; \lambda)$ *to a single coordinate axis),* 107 *squares possess a second exact 3D power-law balance exponent* $\beta^*(S) \in [-3, 4] \setminus \{1\}$ *with* $\mathbf{V}_{\beta^*}(S; 1) = \mathbf{0}$ *(Figure 2(d)).*

3. (Universal positive minimum-variance balancing weights on S^2.) *For every semi-magic square S, the identity $\sum_{i,j} R_{ij}(\lambda)\hat{\mathbf{p}}_{ij}(\lambda) = \sum_{i,j} \mathbf{p}_{ij}(\lambda) = \mathbf{0}$ with $R_{ij}(\lambda) > 0$ places the origin O in the strict relative interior of* $\mathrm{conv}\{\hat{\mathbf{p}}_{ij}(\lambda)\}$. *Letting $\hat{P}(\lambda) \in \mathbb{R}^{N\times 3}$ have rows $\hat{\mathbf{p}}_{ij}(\lambda)^\top$, the unique weight vector $\mathbf{w}^*(S;\lambda)$ of minimum Euclidean distance $\|\mathbf{w}-\mathbf{1}\|_2$ from uniform weights $\mathbf{1}$ satisfying $\sum_{i,j} w^*_{ij}\hat{\mathbf{p}}_{ij}(\lambda) = \mathbf{0}$ is*

$$\mathbf{w}^*(S;\lambda) = \mathbf{1} - \hat{P}(\lambda)\big(\hat{P}(\lambda)^\top \hat{P}(\lambda)\big)^{-1}\mathbf{V}_0(S;\lambda). \tag{13}$$

Across all 880 *order-four magic squares at $\lambda = 1$ (as well as at $\lambda_{\mathrm{iso}} = 1/\sqrt{17}$ and $\lambda_{\mathrm{cube}} = 0.20$), every single square has strictly positive optimal weights $\mathbf{w}^*(S;\lambda) > \mathbf{0}$ (Figure 2(c)), with $\|\mathbf{w}^*(S;1) - \mathbf{1}\|_2 \in [0.00520, 1.5967]$ (mean 0.6583) across the* 832 *non-associative squares.*

Proof. (1) Expanding $\mathbf{V}_3(S;\lambda) = \sum_{i,j}(r_{ij}^2 + \lambda^2 z_{ij}^2)(x_j, y_i, \lambda z_{ij})$, the horizontal sum $\sum_{i,j} r_{ij}^2(x_j, y_i)$ vanishes by odd grid symmetry, leaving $\lambda^2 \sum_{i,j} z_{ij}^2(x_j, y_i)$. The vertical component is $\lambda \sum_{i,j}(x_j^2 + y_i^2)z_{ij} + \lambda^3 \sum_{i,j} z_{ij}^3$; here $\sum_{i,j}(x_j^2 + y_i^2)z_{ij} = 0$ for every semi-magic square because $x_j^2 + y_i^2$ lies in the row–column subspace W_{rc}, and $\sum_{i,j} z_{ij}^3 = 0$ by odd symmetry of the centred entries $\{z_{ij}\}$, proving (12).

(2) For associative squares, $z(-x,-y) = -z(x,y)$ implies $\mathbf{p}_{-c}(\lambda) = -\mathbf{p}_c(\lambda)$ and $R_{-c}(\lambda) = R_c(\lambda)$, so opposite cells cancel pairwise for every $\beta \in \mathbb{R}$ and every even weight function $f(|z_{ij}|)$.

(3) Minimizing $\frac{1}{2}\|\mathbf{w}-\mathbf{1}\|_2^2$ subject to $\hat{P}(\lambda)^\top \mathbf{w} = \mathbf{0}$ by Lagrange multipliers gives (13), and exact enumeration over the 880 squares establishes positivity and the stated extremal bounds. □

Beyond governing static cubic spoke balance, the horizontal vector $\sum_{i,j} z_{ij}^2(x_j, y_i)$ in (12) plays a direct dynamical role when a radial sculpture or Magic Gem is set into three-dimensional rotation. As we show next in Theorem 3.2(3), the two horizontal components of $\mathbf{V}_3(S;\lambda)$ are precisely the vertical off-diagonal products of inertia $(I_{xz}^{(s)}(\lambda), I_{yz}^{(s)}(\lambda))$ of the 3D entry-weighted gem.

3 Torque-Free and Pivot-Top Rotation: Equal Masses Precess Smoothly, Entry Masses Flip

We now turn from static radial equilibrium to three-dimensional rotational dynamics. Suppose a physical Magic Gem (or radial spoke/strut sculpture) of total mass M and centre of mass $\mathbf{c}$ is either tossed in free space (zero gravity) or supported on a low-friction conical pivot or gimbal at the origin O under uniform gravity $\mathbf{g} = (0, 0, -g)^\top$ (generalizing the conical fulcrum of `weighted_ring.stl` in [12]). About the pivot O, gravity exerts the torque

$$\boldsymbol{\tau}_{\mathrm{grav}} = M\,\mathbf{c} \times \mathbf{g},$$

which *vanishes identically in every orientation if and only if the centre of mass lies at the pivot ($\mathbf{c} = \mathbf{0}$)*. Consequently, any balanced sculpture ($\mathbf{c} = \mathbf{0}$) supported at O rotates under Earth's gravity according to Euler's exact torque-free equations [2, 10]: in the principal body frame, the angular velocity vector $\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)^\top$ evolves via

$$I_1\dot{\omega}_1 = (I_2 - I_3)\omega_2\omega_3, \qquad I_2\dot{\omega}_2 = (I_3 - I_1)\omega_3\omega_1, \qquad I_3\dot{\omega}_3 = (I_1 - I_2)\omega_1\omega_2, \tag{14}$$

where I_1, I_2, I_3 are the principal moments of inertia. For an asymmetric body with three distinct principal moments $I_1 < I_2 < I_3$, rotation about the intermediate axis I_2 is exponentially unstable: any small perturbation grows at rate $\sqrt{(I_2 - I_1)(I_3 - I_2)/(I_1 I_3)}\,|\omega_2|$, causing the body to execute the 180° flips of the Dzhanibekov (or tennis-racket) theorem [3, 15].

3.1 Equal Point Masses Make Every Semi-Magic Square an Axisymmetric Rotor with Spherical-Top Scale $\lambda_{\mathrm{iso}} = (n^2+1)^{-1/2}$

Consider first the discrete point cloud $P_S(\lambda) = \{\mathbf{p}_{ij}(\lambda) = (x_j, y_i, \lambda z_{ij})\}$ carrying equal point masses $m = M/N$ at the $N = n^2$ lifted points (for instance, identical steel bearings mounted at the tips of the sixteen radial spokes $\mathbf{p}_{ij}(\lambda)$ on a lightweight hub at O). Because $\mathbf{c} = \frac{1}{N}\sum_{i,j}\mathbf{p}_{ij}(\lambda) = \mathbf{0}$, the gravitational torque about O vanishes identically for every semi-magic square.

Theorem 3.1 (Universal symmetric-top law, spherical-top scale, and periodic polhodes)**.** *Let S be any arrangement of $\{1,\dots,n^2\}$ ($n \geq 2$) whose height field has vanishing first moments $\sum_{i,j} x_j z_{ij} = \sum_{i,j} y_i z_{ij} = 0$ (in particular, any semi-magic or magic square), and place equal point masses $m = M/N$ at the points $\mathbf{p}_{ij}(\lambda) = (x_j, y_i, \lambda z_{ij})$.*

1. (Axisymmetric inertia tensor.) *For every $\lambda > 0$, the inertia tensor about the origin O,*

$$I(\lambda) = \frac{M}{N}\sum_{i,j}\Big(\|\mathbf{p}_{ij}(\lambda)\|_2^2\, I_3 - \mathbf{p}_{ij}(\lambda)\mathbf{p}_{ij}(\lambda)^{\top}\Big),$$

is diagonal and strictly axisymmetric about the z-axis:

$$I(\lambda) = M\,\mathrm{diag}\big(I_\perp(\lambda),\, I_\perp(\lambda),\, I_{zz}\big), \qquad I_\perp(\lambda) = a_n + \lambda^2 b_n, \qquad I_{zz} = 2a_n, \tag{15}$$

with universal principal inertia ratio

$$\frac{I_\perp(\lambda)}{I_{zz}} = \frac{a_n + \lambda^2 b_n}{2a_n} = \frac{1 + \lambda^2(n^2+1)}{2}. \tag{16}$$

2. (Three critical aspect ratios.) *Varying the vertical lift scale λ drives the rotor through three regimes (Figure 4(a)):*

   (a) *At the* canonical lift $\lambda = 1$,

$$\frac{I_\perp(1)}{I_{zz}} = \frac{n^2+2}{2}, \tag{17}$$

   which is an exact integer for every even order n (9 for $n = 4$, 19 for $n = 6$, 33 for $n = 8$) and a half-integer for every odd order n (11/2 for $n = 3$, 27/2 for $n = 5$).

   (b) *At the* spherical-top critical scale

$$\lambda_{\mathrm{iso}}(n) = \frac{1}{\sqrt{n^2+1}} \qquad \Big(\tfrac{1}{\sqrt{10}} \approx 0.3162 \text{ for } n = 3, \quad \tfrac{1}{\sqrt{17}} \approx 0.2425 \text{ for } n = 4\Big), \tag{18}$$

   $I_\perp(\lambda_{\mathrm{iso}}) = I_{zz} = 2a_n$, so $I(\lambda_{\mathrm{iso}}) = 2Ma_nI_3$ is a scalar multiple of the identity: the point-weighted gem is a spherical top *for which every axis through O in $\mathbb{R}^3$ is a principal axis of inertia.*

   (c) *At the* cubic bounding-box scale $\lambda_{\mathrm{cube}}(n) = \frac{n-1}{n^2-1} = \frac{1}{n+1}$ *(where the vertical span matches the horizontal footprint $[-\frac{n-1}{2}, \frac{n-1}{2}]^2$, giving $\lambda_{\mathrm{cube}} = 1/5 = 0.20$ at $n = 4$),*

$$\frac{I_\perp(\lambda_{\mathrm{cube}})}{I_{zz}} = \frac{n^2+n+1}{(n+1)^2} \qquad \Big(\frac{21}{25} = 0.84 \text{ at } n = 4\Big). \tag{19}$$

3. (Suppression of Dzhanibekov tumbling and rational orbit closure.) *Because $I_{xx}(\lambda) = I_{yy}(\lambda) = I_\perp(\lambda)$, no distinct intermediate principal axis exists. In torque-free motion (14), $\omega_3(t) = \omega_{3,0}$ is constant and the transverse angular velocity $(\omega_1(t), \omega_2(t))$ precesses in a uniform circle at the body-frame polhode frequency*

$$\Omega_{\mathrm{body}}(\lambda) = \frac{I_\perp(\lambda) - I_{zz}}{I_\perp(\lambda)}\,\omega_{3,0} = \frac{\lambda^2(n^2+1) - 1}{\lambda^2(n^2+1) + 1}\,\omega_{3,0}. \tag{20}$$

At $\lambda = 1$, $\Omega_{\text{body}}(1) = \frac{n^2}{n^2+2}\omega_{3,0}$ is a rational multiple of the spin rate $\omega_{3,0}$, so the polhode and space-cone trajectories close periodically after $n^2 + 2$ spin revolutions (or $(n^2 + 2)/2$ revolutions when n is even: 9 revolutions at $n = 4$). At $\lambda = \lambda_{\text{iso}}(n)$, $\Omega_{\text{body}} = 0$ and the gem rotates steadily without wobble about any initial angular velocity vector $\boldsymbol{\omega}_0 \in \mathbb{R}^3$.

Proof. By [13, Lem. 3.1], the empirical covariance matrix $\Sigma(\lambda) = \frac{1}{N}\sum_{i,j} \mathbf{p}_{ij}(\lambda)\mathbf{p}_{ij}(\lambda)^\top$ has diagonal entries $\Sigma_{xx} = \Sigma_{yy} = a_n = (n^2 - 1)/12$ and $\Sigma_{zz} = \lambda^2 b_n = \lambda^2(n^4 - 1)/12 = \lambda^2(n^2 + 1)a_n$, while $\Sigma_{xy} = 0$ by grid symmetry and $\Sigma_{xz} = \Sigma_{yz} = 0$ by vanishing of the first moments. Since $I(\lambda) = M\big(\operatorname{tr}(\Sigma(\lambda))I_3 - \Sigma(\lambda)\big)$, we obtain (15) and (16). Specializing (16) to $\lambda = 1$, $\lambda = (n^2+1)^{-1/2}$, and $\lambda = (n + 1)^{-1}$ yields (2a)–(2c). Substituting $I_1 = I_2 = I_\perp(\lambda)$ and $I_3 = I_{zz}$ into Euler's equations (14) gives $\dot\omega_3 = 0$ and $\dot\omega_1 = \Omega_{\text{body}}\omega_2$, $\dot\omega_2 = -\Omega_{\text{body}}\omega_1$ with Ω_{body} as in (20). □

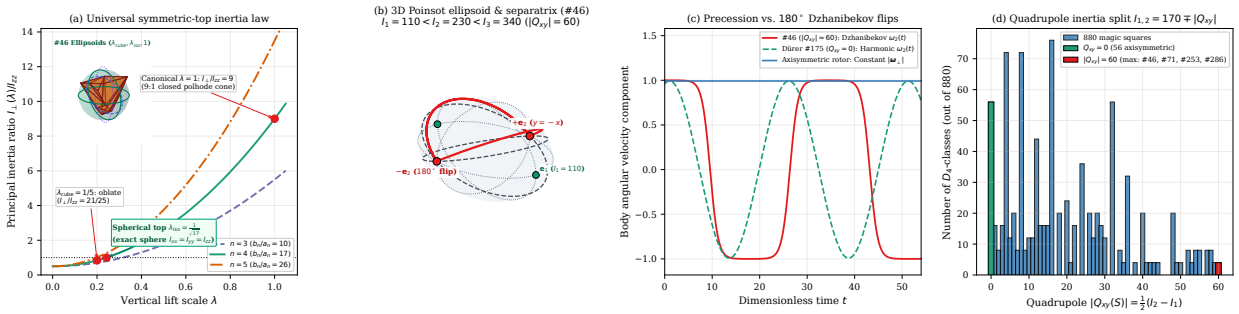


Figure 4: Three-dimensional rotational dynamics of Magic Gems. (a) Universal principal inertia ratio $I_\perp(\lambda)/I_{zz} = \frac{1}{2}(1 + \lambda^2(n^2 + 1))$ versus vertical scale λ for orders $n = 3, 4, 5$, with 3D inertia ellipsoids at the cubic 3D-print scale $\lambda_{\text{cube}} = 1/5$ (21/25), the isotropic spherical-top transition $\lambda_{\text{iso}} = 1/\sqrt{17}$ (1), and the canonical lift $\lambda = 1$ (9) for $n = 4$. (b) Triaxial Poinsot energy ellipsoid $\frac{1}{2}\sum I_k\omega_k^2 = E$ of entry-weighted square #46 ($I_1 = 110 < I_2 = 230 < I_3 = 340$), showing the figure-eight separatrix through the unstable intermediate saddle $\pm\boldsymbol{\omega}_2$ and the heteroclinic polhode orbit connecting $+\omega_2$ to $-\omega_2$. (c) Time series of Euler's equations (14): an axisymmetric magic rotor (Dürer #175 or any point-weighted gem) precesses harmonically with constant $|\boldsymbol{\omega}_\perp|$, whereas entry-weighted square #46 ($|Q_{xy}| = 60$) undergoes periodic 180° Dzhanibekov flips $\omega_2(t) \in [+1, -1]$ about its intermediate diagonal axis. (d) Census of the quadrupole moment $|Q_{xy}(S)| = |\sum z_{ij}x_jy_i| = \frac{1}{2}(I_2 - I_1)$ across the 880 order-four magic squares.

3.2 Entry-Weighted Masses Split the Horizontal Inertia by $2|Q_{xy}|$ and Trigger Dzhanibekov Flips

When each cell of the board or 3D gem instead carries a positive mass $m_{ij} = s_{ij}$ proportional to its magic entry (as in the planar balancing wheel `weighted_ring.stl` [12] or the checker stacks of [13, Fig. 3]), Loly's theorem [11] fixes the diagonal moments of inertia while the off-diagonal products of inertia expose both the planar quadrupole moment $Q_{xy}(S) = \sum_{i,j} z_{ij}x_jy_i$ and the cubic 3D vector resultant $\mathbf{V}_3(S;\lambda)$ of Theorem 2.5(1). Physically, on square #46 (Figure 1(d)), the heavy entries $\{16, 13, 15, 14\}$ concentrate in the NW/SE quadrants ($xy < 0$) while the light entries $\{1, 4, 2, 3\}$ sit in the NE/SW quadrants ($xy > 0$). Holding the board by its opposite NW/SE corners and flicking it to spin about the intermediate diagonal axis $y = -x$ ($I_2 = 230$) places the angular velocity directly on the saddle of the Poinsot energy ellipsoid (Figure 4(b)), triggering a spontaneous 180° mid-air Dzhanibekov flip (Figure 4(c)).

Theorem 3.2 (Entry-weighted inertia split, quadrupole Dzhanibekov instability, and 3D lifted tensor)**.** *Let S be any semi-magic square of order n, and assign entry masses $m_{ij} = s_{ij} = \bar s + z_{ij}$ with total mass $M = \sum_{i,j} s_{ij} = N\bar s = n^2(n^2 + 1)/2$.*

1. (Planar inertia tensor and diagonal principal axes.) *When the masses $m_{ij} = s_{ij}$ are placed at the planar grid positions $(x_j, y_i, 0) \in \mathbb{R}^3$, the centre of mass lies at O and the 3×3 inertia*

tensor about O is

$$I^{(s)}(S) = \begin{pmatrix} I_0 & -Q_{xy}(S) & 0 \\ -Q_{xy}(S) & I_0 & 0 \\ 0 & 0 & 2I_0 \end{pmatrix}, \quad I_0 = Ma_n = \frac{n^2(n^4-1)}{24}, \quad Q_{xy}(S) = \sum_{i,j} z_{ij}x_j y_i. \tag{21}$$

Its principal moments of inertia are

$$I_1(S) = I_0 - |Q_{xy}(S)|, \qquad I_2(S) = I_0 + |Q_{xy}(S)|, \qquad I_3 = 2I_0, \tag{22}$$

with horizontal principal axes directed along the two main diagonals $y = \pm x$.

2. (Order-four Dzhanibekov dichotomy.) *At order $n = 4$, $I_0 = 170$ and $I_3 = 340$. Across the 880 magic squares of order 4 (Figure 4(b–d)):*

   (a) *Exactly 56 squares satisfy $Q_{xy}(S) = 0$ (48 associative squares in Dudeney's Group III, including Dürer's square #175, plus 4 in Group IX and 4 in Group X). For these 56 squares, $I_1 = I_2 = 170$, making the entry-weighted board an axisymmetric rotor with no intermediate-axis instability.*

   (b) *For the remaining 824 squares, $|Q_{xy}(S)|$ takes 47 non-zero integer values in $\{1, \dots, 60\}$ (attaining its maximum $|Q_{xy}| = 60$, $I_1 = 110 < I_2 = 230 < I_3 = 340$, on four squares in Group VI: #46, #71, #253, and #286). Every such square satisfies $I_1 < I_2 < I_3$, so spinning the board about the intermediate diagonal axis $y = -\operatorname{sgn}(Q_{xy})\, x$ triggers the unstable Dzhanibekov flip at dimensionless exponential growth rate*

$$\Gamma(S) = \sqrt{\frac{(I_2 - I_1)(I_3 - I_2)}{I_1 I_3}} = \sqrt{\frac{|Q_{xy}(S)|}{I_0}} = \sqrt{\frac{|Q_{xy}(S)|}{170}}.$$

3. (3D lifted entry-weighted gem, parallel-axis shift, and unification with $\mathbf{V}_3(S;\lambda)$.) *When the entry masses $m_{ij} = s_{ij}$ are placed at the 3D lifted points $\mathbf{p}_{ij}(\lambda) = (x_j, y_i, \lambda z_{ij})$ ($\lambda > 0$), the centre of mass shifts vertically along the z-axis to $\mathbf{c}^{(s)}(\lambda) = (0, 0, z_{\mathrm{cm}}(\lambda))$ with $z_{\mathrm{cm}}(\lambda) = \lambda b_n/\bar{s} = \frac{\lambda(n^2-1)}{6}$ ($= \frac{5}{2}\lambda$ at $n = 4$). Because $\sum_{i,j} z_{ij}^3 = 0$, the diagonal moments of inertia about the fixed grid origin O are*

$$I^{(s)}_{xx,O}(\lambda) = I^{(s)}_{yy,O}(\lambda) = M\big(a_n + \lambda^2 b_n\big), \qquad I^{(s)}_{zz,O}(\lambda) = 2Ma_n = 2I_0$$

($= 170(1 + 17\lambda^2)$ and 340 at $n = 4$), which are identical to those of the equal-mass 3D gem in Theorem 3.1(1). By Steiner's parallel-axis theorem $I^{(s)}_{\mathrm{cm}}(\lambda) = I^{(s)}_O(\lambda) - M\big(\|\mathbf{c}^{(s)}\|_2^2 I_3 - \mathbf{c}^{(s)}\mathbf{c}^{(s)\top}\big)$, translating to the true centre of mass $\mathbf{c}^{(s)}(\lambda)$ for free-space rotation (14) leaves $I^{(s)}_{zz,\mathrm{cm}} = 2I_0$ and all three off-diagonal products of inertia unchanged while subtracting $Mz_{\mathrm{cm}}(\lambda)^2 = M\lambda^2 b_n^2/\bar{s}^2$ ($= 850\lambda^2$ at $n = 4$) from the horizontal diagonal moments:

$$I^{(s)}_{xx,\mathrm{cm}}(\lambda) = I^{(s)}_{yy,\mathrm{cm}}(\lambda) = M\left(a_n + \lambda^2 b_n\left(1 - \frac{b_n}{\bar{s}^2}\right)\right) = Ma_n\left(1 + \frac{2(n^2+2)}{3}\lambda^2\right)$$

($= 170(1 + 12\lambda^2) = 340\big(\frac{1}{2} + 6\lambda^2\big)$ at $n = 4$). In both frames (O and $\mathbf{c}^{(s)}(\lambda)$), the off-diagonal products of inertia are $I^{(s)}_{xy} = -Q_{xy}(S)$ and

$$\big(I^{(s)}_{xz}(\lambda),\, I^{(s)}_{yz}(\lambda)\big) = -\lambda \sum_{i,j} z_{ij}^2 (x_j, y_i) = -\lambda^{-1}\Big(\big(\mathbf{V}_3(S;\lambda)\big)_x,\, \big(\mathbf{V}_3(S;\lambda)\big)_y\Big).$$

Consequently, for all 48 associative order-four magic squares, both $Q_{xy}(S) = 0$ and $\mathbf{V}_3(S;\lambda) = \mathbf{0}$, so the 3D lifted entry-weighted gem has a strictly diagonal, axisymmetric inertia tensor about both O and $\mathbf{c}^{(s)}(\lambda)$.

Proof. (1)–(2) Since S is semi-magic, every row excess $R_i = \sum_j z_{ij}$ and column excess $C_j = \sum_i z_{ij}$ vanishes. Hence $\sum_{i,j} s_{ij}x_j^2 = \bar{s}\sum_{i,j} x_j^2 + \sum_j x_j^2 C_j = Ma_n = I_0$, and similarly $\sum_{i,j} s_{ij}y_i^2 = I_0$, giving $I_{xx} = I_{yy} = I_0$ and $I_{zz} = 2I_0$. Since $\sum_{i,j} x_j y_i = 0$, the off-diagonal product of inertia is $I_{xy} = -\sum_{i,j} s_{ij}x_j y_i = -\sum_{i,j} z_{ij}x_j y_i = -Q_{xy}(S)$. Diagonalizing the 2×2 block $\left(\begin{smallmatrix} I_0 & -Q_{xy} \\ -Q_{xy} & I_0 \end{smallmatrix}\right)$ yields eigenvalues $I_0 \mp |Q_{xy}|$ along $(1, \pm 1)/\sqrt{2}$, proving (21)–(22). At $n = 4$, $I_0 = 16 \cdot \frac{17}{2} \cdot \frac{5}{4} = 170$, and since $|Q_{xy}| \le 60 < 170$, we have $I_1 < I_2 < I_3 = 340$ whenever $Q_{xy} \neq 0$. Because $I_3 - I_2 = 2I_0 - (I_0 + |Q_{xy}|) = I_0 - |Q_{xy}| = I_1$, the factor $I_3 - I_2$ cancels I_1 in the linearized growth rate $\sqrt{(I_2 - I_1)(I_3 - I_2)/(I_1 I_3)} = \sqrt{2|Q_{xy}|I_1/(2I_0 I_1)} = \sqrt{|Q_{xy}|/I_0}$.

(3) At the 3D lifted points $\mathbf{p}_{ij}(\lambda)$, $\sum_{i,j} s_{ij}(x_j, y_i) = \mathbf{0}$ while $\sum_{i,j}(\bar{s} + z_{ij})\lambda z_{ij} = \lambda \sum_{i,j} z_{ij}^2 = N\lambda b_n$, giving $\mathbf{c}^{(s)}(\lambda) = (0, 0, z_{\mathrm{cm}}(\lambda))$ with $z_{\mathrm{cm}}(\lambda) = \lambda b_n/\bar{s}$. Moreover, $\sum_{i,j}(\bar{s} + z_{ij})\lambda^2 z_{ij}^2 = \lambda^2 \bar{s} N b_n + \lambda^2 \sum_{i,j} z_{ij}^3 = M\lambda^2 b_n$ because $\sum_{i,j} z_{ij}^3 = 0$, so $I^{(s)}_{xx,O}(\lambda) = I^{(s)}_{yy,O}(\lambda) = M(a_n + \lambda^2 b_n)$ and $I^{(s)}_{zz,O}(\lambda) = 2Ma_n$. Since $\mathbf{c}^{(s)}(\lambda)$ lies on the z-axis, Steiner's parallel-axis theorem subtracts $Mz_{\mathrm{cm}}(\lambda)^2 \,\mathrm{diag}(1, 1, 0)$ from $I^{(s)}_O(\lambda)$ and leaves all off-diagonal products unchanged; using $b_n = (n^2 + 1)a_n$ and $1 - b_n/\bar{s}^2 = 1 - \frac{n^2-1}{3(n^2+1)} = \frac{2(n^2+2)}{3(n^2+1)}$ gives $b_n(1 - b_n/\bar{s}^2) = \frac{2(n^2+2)}{3}a_n$ (at $n = 4$, $b_4 = 85/4$, $\bar{s} = 17/2$, $1 - b_4/\bar{s}^2 = 12/17$, and $I^{(s)}_{xx,\mathrm{cm}}(\lambda) = 170(1 + 12\lambda^2)$). Finally, $I^{(s)}_{xz}(\lambda) = -\sum_{i,j}(\bar{s} + z_{ij})\lambda x_j z_{ij} = -\lambda \sum_{i,j} z_{ij}^2 x_j = -\lambda^{-1}(\mathbf{V}_3)_x$ and similarly for $I^{(s)}_{yz}(\lambda)$; by Theorem 2.5(1), both vanish alongside $Q_{xy}(S)$ on all 48 associative squares. □

3.3 Uniform Polyhedral Solids Remain Within 2.5% of an Axisymmetric Rotor

When a Magic Gem is fabricated as a uniform-density 3D polyhedron $\mathcal{G}(S) = \mathrm{conv}\, P_S$, its continuous covariance matrix $\Sigma_{\mathrm{solid}} = \frac{1}{\mathrm{vol}\,\mathcal{G}} \iiint_{\mathcal{G}} (\mathbf{x} - \mathbf{c}_0)(\mathbf{x} - \mathbf{c}_0)^\top dV$ and inertia tensor $I_{\mathrm{solid}} = M\big(\mathrm{tr}(\Sigma_{\mathrm{solid}})I_3 - \Sigma_{\mathrm{solid}}\big)$ can be computed in exact rational arithmetic by decomposing $\mathcal{G}(S)$ into tetrahedra (Section A). Across all 880 order-four magic gems at $\lambda = 1$:

- The ratio of the mean transverse principal moment $\frac{1}{2}(I_2 + I_3)$ to the axial moment I_1 (along the near-z axis) has mean 9.412 and range $[7.338, 11.574]$, closely tracking the universal point-mass value 9.
- Because the vanishing of the discrete first moments suppresses the off-diagonal tilt of the hull [13, Sect. 5.3], the relative splitting between the two transverse principal moments remains below 2.5% across the entire order-four census:

$$\frac{I_3 - I_2}{I_2} \in [0.00004,\ 0.0243], \qquad \text{mean } 0.0071.$$

However, while the inertia tensor of the uniform solid $\mathcal{G}(S)$ remains nearly axisymmetric about its own centre of mass $\mathbf{c}_0(S)$, supporting the solid on a fixed pivot at the grid origin O under gravity requires $\mathbf{c}_0(S) = \mathbf{0}$ so that the gravitational torque $\boldsymbol{\tau}_{\mathrm{grav}} = M\mathbf{c}_0(S) \times \mathbf{g}$ vanishes. As we show next in Section 4, passing from the discrete point cloud P_S to the uniform solid $\mathcal{G}_0(S)$ shifts $\mathbf{c}_0(S)$ away from O on 829 of the 880 squares, directly motivating our multi-material core, wireframe, strut-network, and papercraft balancing constructions.

4 Restoring Solid Balance via "Diamond-in-a-Diamond" Convex Layers

A striking mechanical contrast arises as soon as we compare the discrete vector cloud P_S with its solid convex hull $\mathcal{G}_0(S) := \mathcal{G}(S) = \mathrm{conv}\, P_S$. By centring (1), the $N = n^2$ discrete vectors always sum to $\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$. Yet for 829 of the 880 order-four magic squares, the uniform-density solid polyhedron $\mathcal{G}_0(S)$ has a non-zero centre of mass

$$\mathbf{c}_0(S) := \mathbf{c}_{\mathrm{solid}}(\mathcal{G}_0(S)) = \frac{1}{\mathrm{vol}\,\mathcal{G}_0(S)} \iiint_{\mathcal{G}_0(S)} \mathbf{x}\, dV \neq \mathbf{0}.$$

Two geometric mechanisms drive this centroid drift:

1. *Omission of non-vertex points:* The boundary $\partial\mathcal{G}_0(S)$ depends solely on the $V = |\operatorname{vert}\mathcal{G}_0(S)|$ extreme points and ignores the positions of the $N-V$ interior and non-vertex boundary points $P_S^{(1)} = P_S \setminus \operatorname{vert}\mathcal{G}_0(S)$.

2. *Volumetric simplex weighting:* Even when $V = N = 16$ (which occurs for 18 of the 880 order-four gems), a tetrahedral decomposition of $\mathcal{G}_0(S)$ from O weights each simplex Δ_k by its volume $\operatorname{vol}(\Delta_k)$ rather than weighting the 16 vertices equally.

4.1 Fifty-One Order-Four Gems Balance Naturally as Uniform Solids

Proposition 4.1 (Classification of the 51 naturally balanced order-four gems)**.** *Across the* 880 *magic squares of order* 4*, the uniform solid* $\mathcal{G}_0(S)$ *has centre of mass* $\mathbf{c}_0(S) = \mathbf{0}$ *for exactly* 51 *squares:*

1. *The* 48 *associative squares (Dudeney's Group III), for which* $z(-x,-y) = -z(x,y)$ *implies* $\mathcal{G}_0(S) = -\mathcal{G}_0(S)$*, forcing* $\mathbf{c}_0(S) = -\mathbf{c}_0(S) = \mathbf{0}$ *by central inversion.*

2. *Three non-associative squares in Dudeney's Group VI: squares #86, #739, and #756 (Table 1).*

Proof. For every associative square S, $P_S = -P_S$ implies $\mathcal{G}_0(S) = -\mathcal{G}_0(S)$ and $\mathbf{c}_0(S) = \mathbf{0}$. Each of the three Group VI squares #86, #739, and #756 is self-complementary via a reflection ($\overline{S} = hS$ or $\overline{S} = vS$), so $\mathcal{G}_0(S)$ is invariant under a 180° half-turn $\iota_h\colon (x,y,z) \mapsto (x,-y,-z)$ or $\iota_v\colon (x,y,z) \mapsto (-x,y,-z)$ in $\mathbb{R}^3$ [13, Prop. 4.3]. The half-turn forces two coordinates of $\mathbf{c}_0(S)$ to vanish identically ($c_{0,y} = c_{0,z} = 0$ under ι_h, or $c_{0,x} = c_{0,z} = 0$ under ι_v), and exact simplicial integration (Section A) certifies that the single remaining rational axial moment vanishes exclusively on #86, #739, and #756 among all 832 non-associative squares. □

Table 1: The three non-associative order-four magic squares whose uniform solid $\mathcal{G}_0(S)$ has exact centre of mass $\mathbf{c}_0(S) = \mathbf{0}$, together with their inner core $\mathcal{G}_1(S) = \operatorname{conv}(P_S \setminus \operatorname{vert}\mathcal{G}_0(S))$. All three squares lie in Dudeney's Group VI and have 12 outer vertices and 4 inner points (spanning a planar parallelogram in #86 and a 3D tetrahedron in #739 and #756) whose centroid $\mathbf{c}_1(S)$ is also $\mathbf{0}$.

Square	Dudeney	$f(\mathcal{G}_0)$	$\operatorname{vol}(\mathcal{G}_0)$	$\operatorname{Iso}(\mathcal{G}_0)$	$\dim$ / $\operatorname{vol}(\mathcal{G}_1)$	First row of standard form S
#86	Group VI	$(12, 28, 18)$	209/3	C_2 ($\overline{S} = vS$)	2D (0)	$(1, 7, 14, 12)$
#739	Group VI	$(12, 26, 16)$	287/3	C_2 ($\overline{S} = hS$)	3D (5/3)	$(4, 14, 3, 13)$
#756	Group VI	$(12, 24, 14)$	84	C_2 ($\overline{S} = hS$)	3D (4/3)	$(5, 2, 15, 12)$

4.2 Outer Hull Vertices and Inner Core Points Pull in Strictly Opposite Directions

To restore exact physical balance around O for the remaining 829 gems without discarding the omitted grid points, we apply Chazelle's *convex layer decomposition* (or onion peeling) [4] to P_S:

$$P_S^{(0)} = P_S, \qquad \mathcal{G}_k(S) = \operatorname{conv} P_S^{(k)}, \qquad P_S^{(k+1)} = P_S^{(k)} \setminus \operatorname{vert}\mathcal{G}_k(S) \quad (k \geq 0). \tag{23}$$

We call $\mathcal{G}_0(S)$ the *outer gem* and $\mathcal{G}_1(S) = \operatorname{conv} P_S^{(1)} \subset \mathcal{G}_0(S)$ the *inner core gem*—geometrically, "a diamond inside a diamond" (Figure 5(a,b)).

Proposition 4.2 (Vertex-centroid opposition and order-four onion hierarchy)**.** *Let* S *be any arrangement of order* n *with* $V = |\operatorname{vert}\mathcal{G}_0(S)| < N$*.*

1. (Exact vertex-centroid opposition.) *The centroid* $\overline{\mathbf{p}}_{\text{outer}} = \frac{1}{V}\sum_{\mathbf{p}\in\operatorname{vert}\mathcal{G}_0}\mathbf{p}$ *of the outer vertices and the centroid* $\overline{\mathbf{p}}_{\text{inner}} = \frac{1}{N-V}\sum_{\mathbf{p}\in P_S^{(1)}}\mathbf{p}$ *of the inner core points satisfy*

$$V\,\overline{\mathbf{p}}_{\text{outer}} + (N-V)\,\overline{\mathbf{p}}_{\text{inner}} = \mathbf{0}. \tag{24}$$

In particular, $\overline{\mathbf{p}}_{\text{outer}}$ *and* $\overline{\mathbf{p}}_{\text{inner}}$ *are collinear with the origin* O *and point in strictly opposite directions whenever* $\overline{\mathbf{p}}_{\text{outer}} \neq \mathbf{0}$.

2. (Order-four layer census.) *Across the* 880 *magic squares of order* 4 *(Figure 5(c)):*

   - 18 *squares have* $V = 16$ *(*$P_S^{(1)} = \emptyset$*, onion depth* 1*).*
   - 842 *squares have onion depth* 2*, with* $\dim\mathcal{G}_1(S) \in \{0,1,2,3\}$ *distributed as* 14 *singletons (*$V = 15$*),* 162 *collinear segments,* 175 *planar polygons, and* 491 *three-dimensional polyhedra* $\mathcal{G}_1(S)$ *all of whose points are vertices of* $\mathcal{G}_1(S)$*.*
   - 20 *squares have onion depth* 3 *(*$\mathcal{G}_0(S) \supset \mathcal{G}_1(S) \supset \mathcal{G}_2(S)$*), comprising* 16 *squares with dimension sequence* $(3,3,0)$ *(a 3D inner core gem enclosing a single innermost point) and* 4 *squares with dimension sequence* $(3,3,1)$ *(a 3D inner core gem enclosing a 2-point segment).*

   In total, $491 + 16 + 4 = 511$ *order-four magic squares possess a three-dimensional inner core gem* $\mathcal{G}_1(S)$*.*

Proof. Equation (24) follows immediately from $\sum_{\mathbf{p}\in P_S}\mathbf{p} = \mathbf{0}$ and the partition $P_S = \operatorname{vert}\mathcal{G}_0(S) \sqcup P_S^{(1)}$, and exact integer rank and convex-hull peeling across all 880 squares yields the layer distribution in (2). □

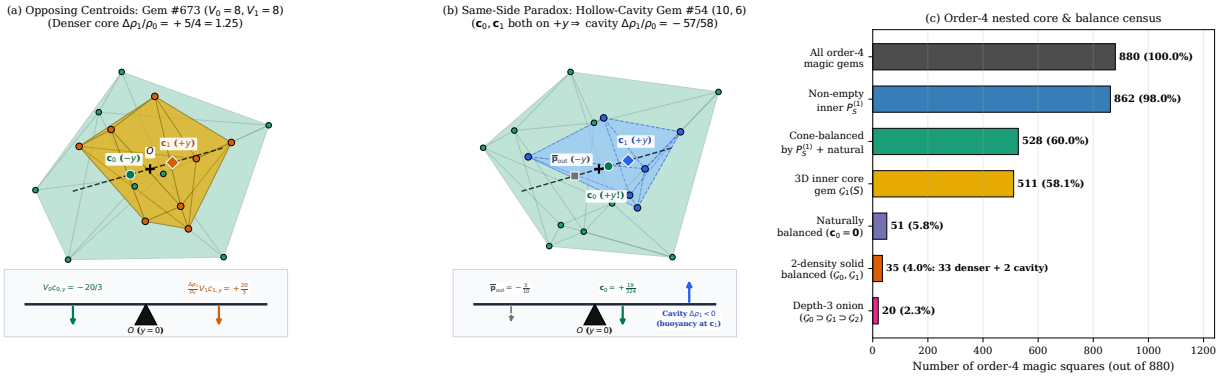


Figure 5: Nested "diamond-in-a-diamond" convex layers $\mathcal{G}_0(S) \supset \mathcal{G}_1(S)$ and centre-of-mass restoration. (a) Denser-core Gem #673 ($(V_0, V_1) = (8,8)$, Figure 1(e)) and (b) hollow-cavity Gem #54 ($(V_0, V_1) = (10,6)$, 1-swap sibling of #45 in Figure 1(c)), showing the translucent outer hull $\mathcal{G}_0(S)$ enclosing the inner core gem $\mathcal{G}_1(S)$ alongside 2D fulcrum lever diagrams at bottom. On #673, $\mathbf{c}_0 = (0,0,-20/257)$ (green circle) and $\mathbf{c}_1 = (0,0,+2/7)$ (gold diamond) lie on opposite sides of O, balancing via a heavier inner core ($\Delta\rho_1/\rho_0 = +5/4$, $\rho_1/\rho_0 = 2.25$). On #54, although the discrete outer vertices still pull to $-y$ ($\overline{\mathbf{p}}_{\text{outer}} = (0,-3/10,0)$), volumetric simplex weighting pulls $\mathbf{c}_0 = (0,+19/324,0)$ onto the $+y$ side alongside $\mathbf{c}_1 = (0,+29/68,0)$, so subtracting a 98.3% hollow core cavity ($\Delta\rho_1/\rho_0 = -57/58$, $\rho_1/\rho_0 = 1/58$) restores balance at O. (c) Breakdown of nested layer properties and balance mechanisms across all 880 order-four magic squares.

4.3 Two-Density Core Inclusions and Inner-Core Weights Restore the Centre of Mass to the Origin

Guided by Proposition 4.2, we establish three physical constructions that fabricate a continuous three-dimensional Magic Gem whose centre of mass lies at O.

Theorem 4.3 (Composite "diamond-in-a-diamond" balancing)**.** *Let S be an order-four magic square.*

1. (Two-density solid inclusion $(\mathcal{G}_0, \mathcal{G}_1)$.) *Suppose* $\dim \mathcal{G}_1(S) = 3$ *(*511 *squares), and fabricate $\mathcal{G}_0(S)$ as a composite solid consisting of a baseline matrix of density $\rho_0 > 0$ throughout $\mathcal{G}_0(S)$ together with an inner core inclusion $\mathcal{G}_1(S) \subset \mathcal{G}_0(S)$ of excess density $\Delta\rho_1 = \rho_1 - \rho_0$. The composite centre of mass is*

$$\mathbf{c}_{\text{comp}} = \frac{\rho_0 \operatorname{vol}(\mathcal{G}_0)\, \mathbf{c}_0 + \Delta\rho_1 \operatorname{vol}(\mathcal{G}_1)\, \mathbf{c}_1}{\rho_0 \operatorname{vol}(\mathcal{G}_0) + \Delta\rho_1 \operatorname{vol}(\mathcal{G}_1)}.$$

Thus $\mathbf{c}_{\text{comp}} = \mathbf{0}$ for some $\Delta\rho_1 \neq 0$ if and only if $\mathbf{c}_0$ and $\mathbf{c}_1$ are non-zero and collinear ($\mathbf{c}_0 \times \mathbf{c}_1 = \mathbf{0}$), with unique balancing density increment

$$\frac{\Delta\rho_1}{\rho_0} = -\frac{\operatorname{vol}(\mathcal{G}_0)\,(\mathbf{c}_0 \cdot \mathbf{c}_1)}{\operatorname{vol}(\mathcal{G}_1)\,\|\mathbf{c}_1\|_2^2}. \tag{25}$$

The 511 *order-four gems with* $\dim \mathcal{G}_1(S) = 3$ *partition into four exact classes:*

- 207 *gems have $\mathbf{c}_1 = \mathbf{0}$ (comprising the* 13 *associative gems with* $\dim \mathcal{G}_1 = 3$ *together with* 194 *non-associative gems—including #739 and #756—whose four inner points form a tetrahedron with vertex sum $\mathbf{0}$);*
- 254 *gems (outside Group VI) have non-collinear solid centroids $\mathbf{c}_0 \times \mathbf{c}_1 \neq \mathbf{0}$;*
- 33 *gems (all in Group VI, where C_2 symmetry locks both $\mathbf{c}_0$ and $\mathbf{c}_1$ onto the symmetry axis) have strictly anti-parallel solid centroids ($\mathbf{c}_0 \cdot \mathbf{c}_1 < 0$), balanced by a* denser *inner core with rational excess density $\Delta\rho_1/\rho_0 \in [5/4, 191/6]$ (Table 2 and Table 5); for instance, on Gem #673 (Figure 1(e), Figure 5(a)), eight extreme entries form a U-shaped outer cradle ($V_0 = 8$, $\overline{\mathbf{p}}_{\text{outer}} = (0, -1/4, 0)$) that swallows the eight intermediate T-core entries ($V_1 = 8$, $\overline{\mathbf{p}}_{\text{inner}} = (0, +1/4, 0)$), and loading $\mathcal{G}_1$ to $\Delta\rho_1/\rho_0 = 5/4$ balances the solid at O;*
- 17 *gems (also in Group VI) have parallel solid centroids pointing in the* same *direction ($\mathbf{c}_0 \cdot \mathbf{c}_1 > 0$), because volumetric simplex weighting of $\mathcal{G}_0(S) \supset \mathcal{G}_1(S)$ pulls $\mathbf{c}_0$ to the same side of O as $\mathbf{c}_1$ despite the opposite discrete vertex centroids of (24). Balancing these* 17 *gems requires $\Delta\rho_1 < 0$, which is physically realizable with non-negative core density $\rho_1 \in [0, \rho_0)$ ($|\Delta\rho_1|/\rho_0 \leq 1$) for two gems: square #123 ($\Delta\rho_1/\rho_0 = -3/17$, $\rho_1/\rho_0 = 14/17 \approx 0.824$, a lighter core) and square #54 ($\Delta\rho_1/\rho_0 = -57/58$, $\rho_1/\rho_0 = 1/58 \approx 0.017$, a nearly hollow core cavity, Figure 5(b)). Notably, square #54 is obtained from denser-core square #45 ($\Delta\rho_1/\rho_0 = +85/61$) by swapping a single complementary pair $(8, 9) \leftrightarrow (15, 2)$ in rows 2–3 (Figure 1(c)): both have $(V_0, V_1) = (10, 6)$, $\overline{\mathbf{p}}_{\text{outer}} = (0, -3/10, 0)$, and $\overline{\mathbf{p}}_{\text{inner}} = (0, +1/2, 0)$, yet inflating the $+y$ hull flares on #54 pulls $\mathbf{c}_0$ across O from $(0, -85/856, 0)$ to $(0, +19/324, 0)$ alongside $\mathbf{c}_1 = (0, +29/68, 0)$.*

2. (Non-negative inner-core point weighting.) *If $\mathcal{G}_0(S)$ has uniform density ρ_0 and non-negative point masses $m_q \geq 0$ are placed at the inner points $q \in P_S^{(1)}$, the composite solid balances at O if and only if $-\mathbf{c}_0(S) \in \operatorname{cone}(P_S^{(1)})$. This cone condition holds for* 477 *of the* 829 *unbalanced gems; combined with the* 51 *naturally balanced gems,* 528 *of the* 880 *order-four gems (*60.0%*) achieve exact balance at O via non-negative inner-core weights.*

3. (Universal barycentric cell / bead-inlaid solid.) *For* every *semi-magic square S of any order n, partitioning $\mathcal{G}_0(S)$ into N cells C_{ij} of equal mass M/N with cell centroids at $\mathbf{p}_{ij}$ (or embedding N identical dense spherical beads of mass m_{bead} at $\mathbf{p}_{ij}$ in a light shell of density $\rho_0 \to 0$) yields $\mathbf{c}_{\text{comp}} = \frac{1}{N}\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$ and restores the exact axisymmetric inertia tensor of Theorem 3.1.*

Proof. Part (1) follows immediately from setting the first moment $\rho_0 \operatorname{vol}(\mathcal{G}_0)\mathbf{c}_0 + \Delta\rho_1 \operatorname{vol}(\mathcal{G}_1)\mathbf{c}_1 = \mathbf{0}$ and evaluating the exact rational simplicial moments (Section A) across all 511 gems with $\dim \mathcal{G}_1(S) = 3$. In Part (2), existence of $m_q \geq 0$ with $\rho_0 \operatorname{vol}(\mathcal{G}_0)\mathbf{c}_0 + \sum_{q \in P_S^{(1)}} m_q \mathbf{p}_q = \mathbf{0}$ is equivalent to $-\mathbf{c}_0 \in \operatorname{cone}(P_S^{(1)})$, certified by linear programming over the 829 unbalanced gems. Part (3) follows from $\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$ and Theorem 3.1. □

Table 2: The eight order-four Magic Gems with the smallest balancing excess density ratios $\Delta\rho_1/\rho_0 \in [5/4, 40/19] \approx [1.25, 2.11]$ in a two-density "diamond-in-a-diamond" solid inclusion $(\mathcal{G}_0(S), \mathcal{G}_1(S))$ (Theorem 4.3(1)), all achievable with standard dual-material 3D printing. All 33 denser-core gems belong to Dudeney's Group VI, where C_2 symmetry forces even vertex counts (V_0, V_1) and aligns $\mathbf{c}_0$ and $\mathbf{c}_1$ along the x- or y-axis. The complete list of 33 gems is given in Table 5 and `anc/results_paper_c.json`.

1-idx	0-idx	(V_0, V_1)	$\operatorname{vol}(\mathcal{G}_0)$	$\operatorname{vol}(\mathcal{G}_1)$	Axial c_0	Axial c_1	Exact $\Delta\rho_1/\rho_0$	Decimal ρ_1/ρ_0
#673	672	(8, 8)	257/3	56/3	−20/257	+2/7	5/4	2.250
#502	501	(10, 6)	271/3	6	+2/271	−1/12	4/3	2.333
#45	44	(10, 6)	214/3	6	−85/856	+61/72	85/61	2.393
#50	49	(10, 6)	254/3	22/3	−7/127	+19/44	28/19	2.474
#307	306	(10, 6)	277/3	22/3	+23/1108	−15/88	23/15	2.533
#49	48	(12, 4)	81	2	−2/81	+1/2	2	3.000
#38	37	(10, 6)	263/3	22/3	+31/1052	−15/88	31/15	3.067
#47	46	(10, 6)	247/3	22/3	−20/247	+19/44	40/19	3.105

From a practical fabrication standpoint, the 33 denser-core solids in Theorem 4.3(1) split into two distinct regimes: the *eight* gems in Table 2 (#673, #502, #45, #50, #307, #49, #38, and #47) require a modest excess density ratio $\Delta\rho_1/\rho_0 \in [5/4, 40/19] \approx [1.25, 2.11]$ (total core-to-shell density ratio $\rho_1/\rho_0 \in [2.25, 3.11]$), which is directly printable on a dual-extruder 3D printer using standard PLA ($\rho_0 \approx 1.24\,\mathrm{g/cm^3}$) around a tungsten- or bronze-filled composite filament core ($\rho_1 \approx 2.8$–$4.0\,\mathrm{g/cm^3}$), whereas the remaining 25 gems in Table 5 have smaller inner-core volumes or shorter lever arms $\|\mathbf{c}_1\|_2$ and require $\Delta\rho_1/\rho_0 \in [47/12, 191/6] \approx [3.92, 31.83]$. Overcoming this solid-density barrier—and extending exact balance to the 194 gems whose tetrahedral core has $\mathbf{c}_{\text{solid}}(\mathcal{G}_1) = \mathbf{0}$—directly motivates the skeletal and thin-shell architectures of Section 4.4: in a nested ball-and-rod wireframe $G(\mathcal{G}_0) \cup G(\mathcal{G}_1)$, linear rod mass density scales quadratically with diameter ($\rho_{\text{lin}} \propto d^2$), compressing the required inner-to-outer rod diameter ratio by a square root ($d_1/d_0 = \sqrt{\rho_{\text{lin},1}/\rho_{\text{lin},0}}$) and balancing 158 Group VI gems using a single homogeneous material.

4.4 Ball-and-Rod Wireframes, Complete K_N Strut Networks, and Unfolded Papercraft Cutout Nets

Beyond solid casting, the companion fabrication suite [12] realizes Magic Gems through three skeletal and thin-shell architectures (Figure 6):

1. *Outer and nested ball-and-rod wireframes* $G(\mathcal{G}_0) \cup G(\mathcal{G}_1)$ (`magicGem_v1.stl`), placing spherical or cubic nodes at the lifted points $\mathbf{p}_u(\lambda)$ and cylindrical rods along the $f_1(\mathcal{G}_0)$ edges of the outer hull $\mathcal{G}_0(S)$ and the $f_1(\mathcal{G}_1)$ edges of the inner core gem $\mathcal{G}_1(S)$ (Figure 6(a));

2. *Fully interconnected complete strut networks* $K_N\big(P_S(\lambda)\big)$ (`magicGem_v2.stl`), connecting all $\binom{N}{2}$ pairs of lifted points $\mathbf{p}_u(\lambda), \mathbf{p}_v(\lambda)$ (120 rods at $n = 4$: $f_1(\mathcal{G}_0)$ outer boundary edges plus $120 - f_1(\mathcal{G}_0)$ internal cross-struts running through the interior of the hull, Figure 6(b)); and

3. *Unfolded 2D papercraft cutout nets with trapezoidal glue tabs*, unfolding the boundary surface $\partial\mathcal{G}_\lambda(S)$ into a flat printable net along a breadth-first spanning tree of the dual facet graph $G^*(\mathcal{G})$ (Figure 6(c)).

Theorem 4.4 (Universal covariance of the complete K_N strut network versus length-weighted and nested wireframes)**.** *Let S be an arrangement of $\{1, \ldots, N\}$, $N = n^2$ $(n \geq 2)$, with lifted point*

cloud $P_S(\lambda) = \{\mathbf{p}_1(\lambda), \dots, \mathbf{p}_N(\lambda)\} \subset \mathbb{R}^3$ *satisfying* $\sum_{u=1}^N \mathbf{p}_u(\lambda) = \mathbf{0}$ *and empirical covariance matrix* $\Sigma(\lambda) = \frac{1}{N}\sum_{u=1}^N \mathbf{p}_u(\lambda)\mathbf{p}_u(\lambda)^\top$. *In the one-dimensional slender-rod (line-skeleton) limit* $d/L_{uv} \to 0^+$, *where each rod is concentrated along its centreline segment:*

1. (Equal-mass-per-edge complete network K_N.) *Suppose each of the* $\binom{N}{2}$ *slender rods connecting* $\mathbf{p}_u(\lambda)$ *and* $\mathbf{p}_v(\lambda)$ $(1 \le u < v \le N)$ *carries equal mass* $m_{\text{edge}} = M_{\text{rods}}/\binom{N}{2}$ *(together with optional equal node masses* $m_{\text{node}} = M_{\text{nodes}}/N$ *at the* N *vertices, where* $M_{\text{nodes}} = \theta M$ *and* $M_{\text{rods}} = (1-\theta)M$ *for* $\theta \in [0,1]$*). Because* K_N *is* $(N-1)$*-regular, the centre of mass of* $K_N\big(P_S(\lambda)\big)$ *lies at the origin* O *for* every *arrangement* S*:*

$$\mathbf{c}\big(K_N(\lambda)\big) = \frac{1}{\binom{N}{2}} \sum_{1\le u<v\le N} \frac{\mathbf{p}_u(\lambda) + \mathbf{p}_v(\lambda)}{2} = \frac{1}{N}\sum_{u=1}^N \mathbf{p}_u(\lambda) = \mathbf{0}. \tag{26}$$

Furthermore, the continuous second-moment covariance matrix of the $\binom{N}{2}$ *equal-mass slender rods is an exact universal scalar multiple of the discrete covariance matrix* $\Sigma(\lambda)$*:*

$$\Sigma_{\text{rods}}^{(=mass)}\big(K_N(\lambda)\big) = \frac{2N-3}{3(N-1)}\,\Sigma(\lambda) \qquad \Big(= \frac{5}{8}\,\Sigma(\lambda) \text{ at } n=3, \quad \frac{29}{45}\,\Sigma(\lambda) \text{ at } n=4\Big). \tag{27}$$

Consequently, for every *semi-magic square* S *of order* n *and every node-mass fraction* $\theta \in [0,1]$*, the composite complete network* $K_N\big(P_S(\lambda)\big)$ *has covariance* $\big(\theta + (1-\theta)\frac{2N-3}{3(N-1)}\big)\Sigma(\lambda)$ *and therefore shares the* exact same axisymmetric inertia ratio $I_\perp(\lambda)/I_{zz} = \frac{1}{2}\big(1+\lambda^2(n^2+1)\big)$, the exact same spherical-top scale $\lambda_{\text{iso}} = (n^2+1)^{-1/2}$, and the exact same integer polhode ratio $(n^2+2)/2 = 9$ at $\lambda = 1$ *as the discrete point cloud in Theorem 3.1.*

2. (Length-weighted strut networks $\rho_{\text{lin}} = \text{const.}$) *When a subgraph* $G = (V_G, E_G)$ *is fabricated from slender rods of uniform cross-sectional diameter* d *and linear mass density* $\rho_{\text{lin}} = \frac{\pi}{4}d^2\rho_{\text{mat}}$*, each rod* $(u,v) \in E_G$ *has mass proportional to its Euclidean length* $L_{uv}(\lambda) = \|\mathbf{p}_u(\lambda) - \mathbf{p}_v(\lambda)\|_2$*, total rod length* $L(G;\lambda) = \sum_{(u,v)\in E_G} L_{uv}(\lambda)$*, and length-weighted centroid*

$$\mathbf{c}_{\text{len}}(G;\lambda) = \frac{1}{2L(G;\lambda)} \sum_{u\in V_G} \left(\sum_{v:\,(u,v)\in E_G} L_{uv}(\lambda) \right) \mathbf{p}_u(\lambda).$$

Across all 880 *order-four magic squares (Figure 6(d)):*

- *Both the* 120*-rod complete network* K_{16} *and the outer hull wireframe* $G(\mathcal{G}_0)$ *have* $\mathbf{c}_{\text{len}} = \mathbf{0}$ *if and only if* S *is one of the* 48 *centrally symmetric associative squares* (5.5%)*.*
- *Passing from the outer hull wireframe* $G(\mathcal{G}_0)$ *(mean total rod length* $\overline{L} = 143.30$ *at* $\lambda = 1$ *and* 60.00 *at* $\lambda = 0.20$*) to the complete* 120*-rod network* K_{16} *(*$\overline{L} = 756.14$ *at* $\lambda = 1$ *and* 305.34 *at* $\lambda = 0.20$*) reduces the mean centroid drift* $\|\mathbf{c}_{\text{len}}\|_2$ *by* 2.7× *at* $\lambda = 1$ *(from* 0.1940 *to* 0.0717*) and by* 3.6× *at* $\lambda = 0.20$ *(from* 0.0879 *to* 0.0247*, versus* 0.0888 *for the uniform solid), while tightening the mean transverse inertia splitting* $(I_3 - I_2)/I_2$ *to* 0.0051 *at* $\lambda = 1$*.*

3. (Composite balance of nested wireframes $G(\mathcal{G}_0) \cup G(\mathcal{G}_1)$ via rod diameter tuning.) *For the* 848 *order-four gems with* $\dim \mathcal{G}_1(S) \ge 1$*, fabricating the outer edges* $E(\mathcal{G}_0)$ *with rod diameter* d_0 *and the disjoint inner core edges* $E(\mathcal{G}_1)$ *with rod diameter* d_1 *yields* $\mathbf{c}_{\text{wire}}(\mathcal{G}_0 \cup \mathcal{G}_1) = \mathbf{0}$ *whenever* $\mathbf{c}_{\text{wire}}(\mathcal{G}_0)$ *and* $\mathbf{c}_{\text{wire}}(\mathcal{G}_1)$ *are strictly anti-parallel, at the unique rod diameter ratio*

$$\frac{d_1}{d_0} = \sqrt{\frac{\rho_{\text{lin},1}}{\rho_{\text{lin},0}}} = \sqrt{-\frac{L\big(G(\mathcal{G}_0)\big)\,\big(\mathbf{c}_{\text{wire}}(\mathcal{G}_0)\cdot\mathbf{c}_{\text{wire}}(\mathcal{G}_1)\big)}{L\big(G(\mathcal{G}_1)\big)\,\|\mathbf{c}_{\text{wire}}(\mathcal{G}_1)\|_2^2}}. \tag{28}$$

Because $E(\mathcal{G}_0) \cap E(\mathcal{G}_1) = \emptyset$*, any ratio* $d_1/d_0 > 0$ *is physically realizable without cavity constraints. Consequently, exact composite wireframe balance holds for* 158 *Group VI gems*

at $\lambda = 1$ (81 with $\dim \mathcal{G}_1 = 3$, 37 with $\dim \mathcal{G}_1 = 2$, 40 with $\dim \mathcal{G}_1 = 1$; e.g., Gem #673 at $d_1/d_0 = 0.7932$ and Gem #45 at $d_1/d_0 = 0.7945$) and for 164 Group VI gems at $\lambda = 0.20$ (83 with $\dim \mathcal{G}_1 = 3$; e.g., Gem #673 at $d_1/d_0 = 0.4932$ and Gem #45 at $d_1/d_0 = 0.7694$).

Proof. (1) Integrating $\mathbf{x}(t) = (1-t)\mathbf{p}_u + t\mathbf{p}_v$ over $t \in [0,1]$ gives the rod midpoint $\frac{1}{2}(\mathbf{p}_u + \mathbf{p}_v)$ and second-moment matrix

$$M_2(u,v) = \int_0^1 \mathbf{x}(t)\mathbf{x}(t)^\top \, dt = \frac{1}{3}\big(\mathbf{p}_u\mathbf{p}_u^\top + \mathbf{p}_v\mathbf{p}_v^\top\big) + \frac{1}{6}\big(\mathbf{p}_u\mathbf{p}_v^\top + \mathbf{p}_v\mathbf{p}_u^\top\big).$$

In K_N, each vertex u belongs to $N-1$ edges, proving (26) and giving $\sum_{u<v}(\mathbf{p}_u\mathbf{p}_u^\top + \mathbf{p}_v\mathbf{p}_v^\top) = (N-1)\sum_{u=1}^N \mathbf{p}_u\mathbf{p}_u^\top = N(N-1)\Sigma(\lambda)$. Similarly, expanding $\mathbf{0} = \big(\sum_u \mathbf{p}_u\big)\big(\sum_v \mathbf{p}_v\big)^\top$ yields $\sum_{u<v}(\mathbf{p}_u\mathbf{p}_v^\top + \mathbf{p}_v\mathbf{p}_u^\top) = -\sum_{u=1}^N \mathbf{p}_u\mathbf{p}_u^\top = -N\Sigma(\lambda)$. Dividing by $\binom{N}{2} = \frac{1}{2}N(N-1)$ gives

$$\Sigma_{\text{rods}}^{(=\text{mass})}\big(K_N(\lambda)\big) = \frac{2}{N(N-1)}\left(\frac{N-1}{3} - \frac{1}{6}\right) N\Sigma(\lambda) = \frac{2N-3}{3(N-1)}\,\Sigma(\lambda),$$

establishing (27). Parts (2) and (3) follow by evaluating $L(G;\lambda)$, $\mathbf{c}_{\text{len}}(G;\lambda)$, and (28) across all 880 squares (Section A). □

Proposition 4.5 (Papercraft cutout nets, topological glue-tab invariant, and thin-shell balance)**.** *Let $\mathcal{G}_\lambda(S)$ be any three-dimensional Magic Gem with f-vector (f_0, f_1, f_2).*

1. (Universal glue-tab invariant $N_{\text{tabs}} = f_0 - 1$ and non-overlapping order-four nets.) *Unfolding $\partial\mathcal{G}_\lambda(S)$ into a connected planar cutout net along any spanning tree T^* of the dual facet graph $G^*(\mathcal{G})$ [7, 18, 5, 9] partitions the f_1 polyhedral edges into $|E_{\text{fold}}| = f_2 - 1$ interior fold lines (dashed creases in Figure 6(c)) and $N_{\text{tabs}} = f_1 - (f_2 - 1)$ cut edges, each opened into a pair of exterior boundary segments joined by one trapezoidal glue tab (labelled in matching pairs $\mathtt{a}$–$\mathtt{k}$ in Figure 6(c)). Because $\partial\mathcal{G}_\lambda(S) \cong S^2$ is a genus-0 planar surface, the cut edges $E \setminus E(T^*)$ form a complementary spanning tree T of the primal 1-skeleton $G(\mathcal{G}_\lambda)$ on the f_0 hull vertices (equivalently, the cycle rank $\beta_1(G^*) = f_1 - f_2 + 1$ of the dual graph equals $f_0 - 1$ by Euler's formula $f_0 - f_1 + f_2 = 2$), so the number of glue tabs depends* solely on the hull vertex count f_0*:*

$$N_{\text{tabs}} = f_1 - (f_2 - 1) = |E(T)| = f_0 - 1, \tag{29}$$

and the net perimeter consists of $2(f_0 - 1)$ cut boundary segments. Moreover, (29) is strictly invariant under triangulating non-triangular facets into $F_\Delta = 2f_0 - 4$ triangles and $E_\Delta = 3f_0 - 6$ edges, since $E_\Delta - (F_\Delta - 1) = (3f_0 - 6) - (2f_0 - 5) = f_0 - 1$. Across the 880 order-four Magic Gems, $N_{\text{tabs}} = f_0 - 1$ takes every integer value in $\{7, \ldots, 15\}$ (distributed as 7 gems with 7 tabs, 10 with 8, 103 with 9, 80 with 10, 380 with 11 including Dürer's gem #175, 106 with 12, 162 with 13, 14 with 14, and 18 with 15 tabs). Under the deterministic breadth-first dual spanning tree T^* rooted at the largest-area facet F_0 (marked $\bigstar F_0$ in Figure 6(c), visiting adjacent facets in canonical index order), all 880 order-four Magic Gems (100%) unfold into strictly non-overlapping 2D planar nets *at both $\lambda = 1$ and $\lambda = 0.20$.*

2. (Hollow thin-shell mechanics $\partial\mathcal{G}_\lambda(S)$ and nested shell balance.) *In the idealized mathematical shell model $\partial\mathcal{G}_\lambda(S)$ of uniform areal mass density $\sigma_0 > 0$ (with massless glue tabs, or butt-jointed laser-cut facets), the surface area is* $\text{area}(\partial\mathcal{G}_\lambda) \in [116.53, 148.62]$ *(mean 135.13) at $\lambda = 1$ and $[31.26, 40.02]$ (mean 36.38) at $\lambda = 0.20$, with area-weighted centroid $\mathbf{c}_{\text{shell}}(\mathcal{G}_0) = \frac{1}{\text{area}(\partial\mathcal{G}_0)}\sum_F A_F\mathbf{c}(F)$ vanishing on all 48 associative squares and having mean drift 0.1357 at $\lambda = 1$ (0.0907 at $\lambda = 0.20$). Enclosing a nested inner shell $\partial\mathcal{G}_1(S)$ ($\dim \mathcal{G}_1 = 3$) of areal density σ_1 restores exact composite balance $\mathbf{c}_{\text{shell}}(\partial\mathcal{G}_0 \cup \partial\mathcal{G}_1) = \mathbf{0}$ for 80 Group VI gems at $\lambda = 1$ and 87 Group VI gems at $\lambda = 0.20$ (including Gem #673 at $\sigma_1/\sigma_0 = 0.9110$ for $\lambda = 1$ and 0.6826 for $\lambda = 0.20$, and Gem #45 at $\sigma_1/\sigma_0 = 0.0849$ for $\lambda = 1$ and 0.2003 for $\lambda = 0.20$).*

Proof. By planar graph duality on $\partial\mathcal{G}_\lambda(S) \cong S^2$, the complement of a dual spanning tree $T^* \subset G^*(\mathcal{G}_\lambda)$ ($|E(T^*)| = f_2 - 1$) is a primal spanning tree $T \subset G(\mathcal{G}_\lambda)$ on the f_0 vertices, giving $N_{\text{tabs}} = |E(T)| = f_0 - 1$ directly (and matching $f_1 - (f_2 - 1) = f_0 - 1$ by Euler's formula). Whether every convex polyhedron admits a non-overlapping edge unfolding remains open in general (Dürer's conjecture [7, 18, 5, 9]); for the 880 order-four Magic Gems at $\lambda = 1$ and $\lambda = 0.20$, exact 2D polygon intersection testing of the breadth-first dual spanning-tree unfolding and exact triangular surface moments certify non-overlap and the stated shell censuses across all 880 gems (Section A). □

Remark 4.6 (Finite-diameter cylinders, hub intersections, and core suspension). In a physical 3D print of finite rod diameter $d > 0$, each right circular cylinder of length L_{uv} and unit direction $\hat{\mathbf{e}}_{uv} = (\mathbf{p}_v - \mathbf{p}_u)/L_{uv}$ has midpoint centroid $\frac{1}{2}(\mathbf{p}_u + \mathbf{p}_v)$ (leaving (26) unchanged) and adds the transverse radial inertia correction $m_{uv}\frac{d^2}{16}(I_3 + \hat{\mathbf{e}}_{uv}\hat{\mathbf{e}}_{uv}^\top)$ to its slender-rod inertia tensor, while boolean union at the 16 degree-15 vertex hubs and interior strut crossings subtracts $O(d^3)$ cap-overlap volumes. To suspend the disjoint inner core wireframe $G(\mathcal{G}_1)$ inside $G(\mathcal{G}_0)$ on a Group VI gem, one connects them by either lightweight tension filaments or C_2-symmetric pairs of equal-length struts whose midpoint centroid lies at O, leaving the balancing diameter ratio d_1/d_0 in (28) invariant. Finally, the jump from 35 two-density balanced solids (Theorem 4.3(1)) to 158 balanced nested wireframes at $\lambda = 1$ arises because 194 non-associative gems have a 4-vertex tetrahedral core $\mathcal{G}_1(S)$ with $\sum_{q \in P_S^{(1)}} \mathbf{p}_q = \mathbf{0}$: its solid centroid is $\mathbf{c}_{\text{solid}}(\mathcal{G}_1) = \frac{1}{4}\sum \mathbf{p}_q = \mathbf{0}$, which cannot counter $\mathbf{c}_0 \neq \mathbf{0}$, whereas its six edges have unequal Euclidean lengths on opposite sides of O, producing a non-zero wireframe centroid $\mathbf{c}_{\text{wire}}(\mathcal{G}_1) \neq \mathbf{0}$ along the Group VI C_2 axis.

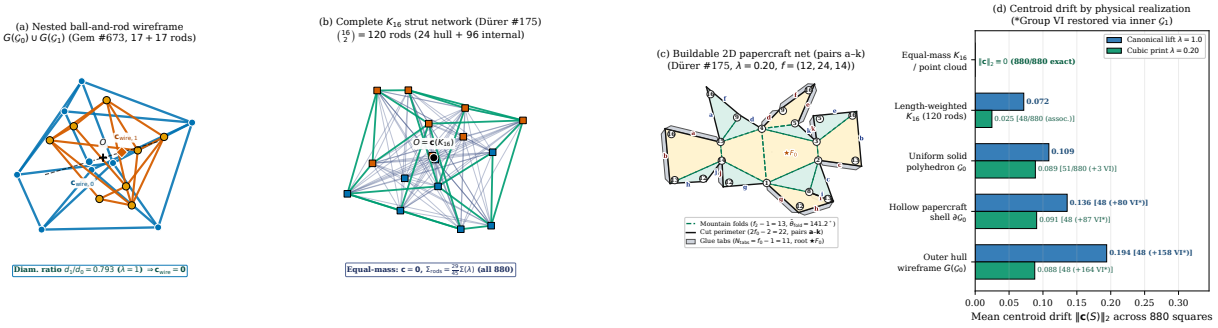


Figure 6: Wireframes, complete K_{16} strut networks, and unfolded papercraft cutout nets of Magic Gems [12]. (a) Nested ball-and-rod wireframe $G(\mathcal{G}_0) \cup G(\mathcal{G}_1)$ of Gem #673 (17 outer blue rods + 17 inner orange rods), where tuning the inner-to-outer rod diameter ratio to $d_1/d_0 = 0.793$ at $\lambda = 1$ (0.493 at $\lambda = 0.20$) restores $\mathbf{c}_{\text{wire}} = \mathbf{0}$. (b) Fully interconnected complete network K_{16} of Dürer's Gem #175 (`magicGem_v2.stl`) with all $\binom{16}{2} = 120$ rods (24 outer hull edges in green + 96 internal cross-struts in slate) and 16 node cubes; for equal-mass rods, $\mathbf{c}(K_{16}) = \mathbf{0}$ and $\Sigma_{\text{rods}} = \frac{29}{45}\Sigma(\lambda)$ for all 880 squares (Theorem 4.4(1)). (c) Exact 2D unfolded papercraft cutout net of Dürer's Gem #175 ($\lambda = 0.20$, $f = (12, 24, 14)$) rooted at $\bigstar F_0$, with $f_2 - 1 = 13$ dashed fold lines (mean dihedral fold angle $\overline{\theta}_{\text{fold}} = 141.2° \in [117°, 165°]$), $2f_0 - 2 = 22$ solid cut perimeter segments, $N_{\text{tabs}} = f_0 - 1 = 11$ grey trapezoidal glue tabs paired with their mating cut edges by lowercase labels `a`–`k` (Proposition 4.5(1)), and magic entries s_{ij} at the unfolded vertices. (d) Comparison of mean centroid drift $\|\mathbf{c}(S)\|_2$ across all 880 order-four magic squares for the five physical realizations at $\lambda = 1.0$ and $\lambda = 0.20$.

5 Mirror-Twin Inversion Locks Gems into Vertical Towers

Our final mechanical investigation treats Magic Gems as physical building blocks under gravity. When a polyhedron $\mathcal{G}_\lambda(S)$ rests on a flat horizontal table on one of its facets F, or when multiple gems are stacked face-to-face into a vertical column, two geometric questions arise: (i) when is a resting facet statically stable against tipping under gravity, and (ii) how can copies of a gem be stacked along slanted facets without the column curving away from the vertical and toppling?

5.1 Cubic Aspect Scaling Raises Facet Resting Stability from 43.0% to 74.5%

Let $F \subset \partial\mathcal{G}_\lambda(S)$ be a facet with outward unit normal $\hat{\mathbf{n}}_F(\lambda)$. When $\mathcal{G}_\lambda(S)$ rests with F on a horizontal plane (so $\hat{\mathbf{n}}_F(\lambda)$ points straight down), gravity acts vertically through the centre of mass $\mathbf{c}(\lambda)$. Let $\pi_F(\mathbf{c}(\lambda))$ denote the orthogonal projection of $\mathbf{c}(\lambda)$ onto the affine plane of F, and let $h_F(\lambda) = (\mathbf{p}_F - \mathbf{c}(\lambda)) \cdot (-\hat{\mathbf{n}}_F(\lambda)) > 0$ be the height of the centre of mass above the table.

Definition 5.1 (Static stability margin and critical tipping angle)**.** For a facet $F \subset \partial\mathcal{G}_\lambda(S)$, the *stability margin* $d_{\text{stab}}(S, F; \lambda)$ is the signed Euclidean distance in the plane of F from $\pi_F(\mathbf{c}(\lambda))$ to the boundary ∂F of the facet polygon (taken positive when $\pi_F(\mathbf{c}(\lambda))$ lies in the strict relative interior of F, and negative outside). Facet F is *statically stable* if and only if $d_{\text{stab}}(S, F; \lambda) > 0$, in which case its *critical tipping angle* (the maximum tilt of the table before $\mathcal{G}_\lambda(S)$ topples across an edge of F) is

$$\theta_{\text{tip}}(S, F; \lambda) = \arctan\left(\frac{d_{\text{stab}}(S, F; \lambda)}{h_F(\lambda)}\right). \tag{30}$$

Proposition 5.2 (Order-four resting stability census)**.** *Across all* 880 *magic squares of order* 4*, the gems* $\mathcal{G}_\lambda(S)$ *possess a total of* 14,824 *facets (mean* 16.85 *facets per gem).*

1. *At the canonical lift* $\lambda = 1$*, using the exact uniform-solid centre of mass* $\mathbf{c}_0(S)$*, exactly* 6,378 *of the* 14,824 *facets (*43.0%*) are statically stable, with every gem possessing between* 4 *and* 12 *stable resting facets (mean* 7.25*; using the balanced point centroid* O *gives* 6,340 *stable facets,* 42.8%*).*

2. *At the cubic 3D-print scale* $\lambda_{\text{cube}} = 1/5 = 0.20$ *(where the height span of* $1, \ldots, 16$ *equals the* 3×3 *grid footprint), the reduced vertical aspect ratio brings the centre of mass closer to the slanted upper and lower facets:* 11,044 *of the* 14,824 *facets (*74.5%*) are statically stable, with a mean critical tipping angle of* 17.8° *and a maximum of* 47.3° *(Figure 7(d)).*

Proof. For each of the 880 gems and each facet $F \subset \partial\mathcal{G}_\lambda(S)$ with exact integer plane equation $aX + bY + cZ + d = 0$ in doubled coordinates, rescaling $z \mapsto \lambda z$ transforms the outward normal to $\mathbf{n}_F(\lambda) \propto (a\lambda, b\lambda, c)$ and the solid centroid to $\mathbf{c}_0(\lambda) = (c_{0,x}, c_{0,y}, \lambda c_{0,z})$. Projecting $\mathbf{c}_0(\lambda)$ onto an orthonormal basis of the affine plane of F and evaluating the signed edge distances $d_{\text{stab}}(S, F; \lambda)$ and (30) across all 14,824 facets at $\lambda = 1$ and $\lambda = 0.20$ yields the stated counts and extremal angles. □

5.2 Central Inversion Pairs Every Upper Facet with a Parallel Lower Facet of Its Mirror Twin

Suppose we wish to stack multiple blocks vertically along an upper facet $F_+ \subset \mathcal{G}_\lambda(S)$ (with outward normal $\mathbf{n}_+$, $n_{+,z} > 0$) and a lower facet $F_- \subset \mathcal{G}_\lambda(S)$ (with outward normal $\mathbf{n}_-$, $n_{-,z} < 0$). Stacking translated copies of the *same* gem $\mathcal{G}_\lambda(S)$ without rotation requires $\mathbf{n}_- = -\mathbf{n}_+$ (which holds for only 124 of the 880 gems); when $\mathbf{n}_- \neq -\mathbf{n}_+$, same-gem copies meet at an open wedge gap (Figure 7(c), left). Pairing a gem with its *rotational complement (mirror twin)*

$$S^* := r^2\overline{S} = (N+1)\mathbf{1} - r^2 S \tag{31}$$

solves the stacking problem universally via central inversion in $\mathbb{R}^3$ (Figure 7(a,c)).

Theorem 5.3 (Mirror-Twin Inversion and Zero-Rotation Alternating Towers)**.** *Let* S *be any arrangement of order* n*, let* $S^* = r^2\overline{S}$ *be its half-turned complement, and let* $\lambda > 0$*.*

1. (Central inversion identity.) *In* $\mathbb{R}^3$*, the lifted point clouds and convex hulls satisfy*

$$P_{S^*}(\lambda) = -P_S(\lambda), \qquad \mathcal{G}_\lambda(S^*) = -\mathcal{G}_\lambda(S). \tag{32}$$

Consequently, the map $F \mapsto F^* := -F$ *is an area-preserving, normal-reversing bijection between the upper facets* $F_+ \subset \mathcal{G}_\lambda(S)$ *(outward normal* $+\mathbf{n}_+$*) and the lower facets* $F_+^* \subset \mathcal{G}_\lambda(S^*)$ *(outward normal* $-\mathbf{n}_+$*), and between the lower facets* $F_- \subset \mathcal{G}_\lambda(S)$ *(outward normal* $+\mathbf{n}_-$*) and the upper facets* $F_-^* = -F_- \subset \mathcal{G}_\lambda(S^*)$ *(outward normal* $-\mathbf{n}_-$*).*

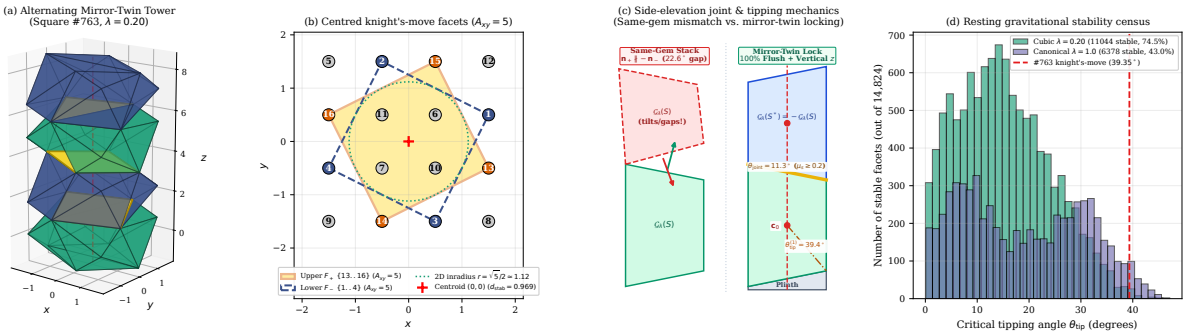


Figure 7: Mirror-twin facet locking and periodic vertical towers. (a) Four-block alternating tower of square #763 at the cubic 3D-print scale $\lambda = 0.20$, alternating $\mathcal{G}_\lambda(S)$ (green) with its mirror twin $\mathcal{G}_\lambda(r^2\overline{S}) = -\mathcal{G}_\lambda(S)$ (blue); the golden contact joints are the centred knight's-move squares $\{13, 14, 15, 16\}$ and $\{1, 2, 3, 4\}$, which meet with 100% vertex congruence and keep the column's centre of mass on the vertical z-axis (red dashed). (b) Top-down (x, y) footprint of square #763 showing the dual knight's-move squares of area $A_{xy} = 5$ and 2D footprint inradius $r = \sqrt{5}/2 \approx 1.12$ (while the 3D tilted-plane stability margin from $\mathbf{c}_0$ is $d_{\text{stab}} = 0.969$ at $\lambda = 0.20$). (c) Two-dimensional side-elevation joint and tipping schematic comparing the $2\theta_{\text{joint}} = 22.6°$ wedge mismatch of unrotated same-gem stacking against flush mirror-twin locking ($\mathbf{n}_+^* = -\mathbf{n}_+$) on a $\theta_{\text{joint}} = 11.3°$ base plinth. (d) Census of critical tipping angles θ_{tip} over all 14,824 facets at $\lambda = 0.20$ and $\lambda = 1.0$.

2. (Zero-rotation alternating tower.) *Choose* any *upper facet* $F_+ \subset \mathcal{G}_\lambda(S)$ *and* any *lower facet* $F_- \subset \mathcal{G}_\lambda(S)$ *(not necessarily parallel) that are centrally symmetric polygons in* $\mathbb{R}^3$ *with centroids* $\mathbf{c}(F_+)$ *and* $\mathbf{c}(F_-)$. *Define the alternating sequence of unrotated blocks* $B_0, B_1, B_2, \ldots$ *by*

$$B_{2m} = \mathcal{G}_\lambda(S) + m\mathbf{T}, \qquad B_{2m+1} = -\mathcal{G}_\lambda(S) + 2\mathbf{c}(F_+) + m\mathbf{T}, \qquad \mathbf{T} = 2\big(\mathbf{c}(F_+) - \mathbf{c}(F_-)\big). \quad (33)$$

Then:

   (a) *At every odd joint* $B_{2m} \cap B_{2m+1}$, *the upper facet* $F_+ + m\mathbf{T}$ *of* B_{2m} *coincides* identically *(100% polygon and vertex overlap, with opposite outward normals* $\pm\mathbf{n}_+$*) with the lower facet* $-F_+ + 2\mathbf{c}(F_+) + m\mathbf{T}$ *of* B_{2m+1}.

   (b) *At every even joint* $B_{2m-1} \cap B_{2m}$, *the upper facet* $-F_- + 2\mathbf{c}(F_+) + (m-1)\mathbf{T}$ *of* B_{2m-1} *coincides* identically *(with opposite outward normals* $\mp\mathbf{n}_-$*) with the lower facet* $F_- + m\mathbf{T}$ *of* B_{2m}.

   (c) *Every two-block repeating unit* $B_{2m} \cup B_{2m+1}$ *is centrally symmetric about* $\mathbf{c}(F_+) + m\mathbf{T}$. *When* $\mathbf{c}(F_+) = (0, 0, z_+)$ *and* $\mathbf{c}(F_-) = (0, 0, z_-)$ *both lie on the vertical* z*-axis* ($z_+ > 0 > z_-$), *the period vector* $\mathbf{T} = (0, 0, 2(z_+ - z_-))$ *is purely vertical, and the centre of mass of any* $2M$*-block tower lies on the* z*-axis regardless of whether* $\mathbf{c}_0(S) = \mathbf{0}$ *(Figure 7(a,c)).*

Proof. (1) In centred coordinates (1), r^2 sends $(x, y) \mapsto (-x, -y)$ and complementation sends $z \mapsto -z$, so $r^2\overline{S}$ lifts to $-\mathbf{p}_{ij}(\lambda)$, proving (32).

(2) Central symmetry of $F_\pm$ about $\mathbf{c}(F_\pm)$ gives $-F_\pm + 2\mathbf{c}(F_\pm) = F_\pm$. Translating by $m\mathbf{T}$ yields (2a) and (2b), while central inversion about $\mathbf{c}(F_+) + m\mathbf{T}$ exchanges B_{2m} and B_{2m+1}, proving (2c). □

5.3 Seventeen Gems Form Area-5 Towers via Centred Knight's-Move Square Facets

Proposition 5.4 (Vertically centered tower census and the knight's-move square #763)**.** *Across the* 880 *magic squares of order* 4*:*

1. *Exactly* 48 *squares possess both an upper facet* F_+ *and a lower facet* F_- *that are centrally symmetric polygons whose centroids lie on the vertical axis* $(0, 0, z_\pm)$.

2. *The maximum projected horizontal contact area among these* 48 *squares is* $A_{xy} = 5$ *(spanning* $5/9 \approx 55.6\%$ *of the* 3×3 *grid footprint), achieved by exactly* 17 *squares (listed in Table 3; e.g. squares #603, #763, and #774). In all* 17 *squares, the projected polygon* $\pi_{xy}(F_+)$ *is the centred* 2×1 *knight's-move square with vertices* $(\pm\frac{3}{2}, \mp\frac{1}{2})$ *and* $(\pm\frac{1}{2}, \pm\frac{3}{2})$ *(Figure 7(b)).*

3. *Of these* 17 *knight's-move gems, only two (#774 and #792) have parallel top and bottom facets* ($\mathbf{n}_- = -\mathbf{n}_+$). *The remaining* 15 *gems cannot form a vertical column from unrotated copies of a single gem, but lock into a purely vertical, zero-drift column when alternated with their mirror twin.*

4. *In particular, square #763 (Dudeney Group VI,* $\overline{S} = hS$, *so* $S^* = r^2\overline{S} = vS$ *is its vertical reflection) is obtained from naturally balanced solid #756 (*$\mathbf{c}_0 = \mathbf{0}$, *Table 1) by swapping the single complementary pair* $(4, 13) \leftrightarrow (16, 1)$ *in rows* 2–3 *(Figure 1(f)):*

$$S_{\#763} = \begin{pmatrix} 5 & 2 & \mathbf{15} & 12 \\ \mathbf{16} & 11 & 6 & 1 \\ 4 & 7 & 10 & \mathbf{13} \\ 9 & \mathbf{14} & 3 & 8 \end{pmatrix}, \qquad S^*_{\#763} = vS_{\#763} = \begin{pmatrix} 9 & 14 & 3 & 8 \\ 4 & 7 & 10 & 13 \\ 16 & 11 & 6 & 1 \\ 5 & 2 & 15 & 12 \end{pmatrix}.$$

This single swap lifts $\{13, 14, 15, 16\}$ *onto the Pythagorean upper knight's-move plane* $4x - 3y + 5z = 30$ *(centroid* $(0, 0, 6\lambda)$*) and* $\{1, 2, 3, 4\}$ *onto the lower plane* $-4x - 3y - 5z = 30$ *(centroid* $(0, 0, -6\lambda)$*), while shifting the solid centroid slightly to* $\mathbf{c}_0 = (0, -5/126, 0)$ *(which cancels pairwise in* $B_{2m} \cup B_{2m+1}$*). Both facets have 3D area* $5\sqrt{1+\lambda^2}$ *(*$\sqrt{26} \approx 5.099$ *at* $\lambda = 0.20$*), stability margin* $d_{\text{stab}} = 0.9694$, *and single-block tipping angle* $\theta^{(1)}_{\text{tip}} = 39.35°$ *at* $\lambda = 0.20$. *Because* $\mathbf{n}_\pm(\lambda) \propto (\pm 4\lambda, -3\lambda, \pm 5)$ *makes angle* $\theta_{\text{joint}}(\lambda) = \arctan(\lambda)$ *with* $\hat{\mathbf{e}}_z$ *(*$\sqrt{4^2 + (-3)^2} = 5$*), scaling from* $\lambda = 1$ *(*$\theta_{\text{joint}} = 45°$*) to* $\lambda = 0.20$ *reduces the joint slope to* $\arctan(0.20) \approx 11.31°$*: any material with static friction* $\mu_s > \lambda = 0.20$ *holds the unbonded column at level (*$\theta_{\text{table}} = 0$*), while tilting an* M*-block column of centre-of-mass height* H_M *is governed by* $\min\big(\arctan(\mu_s) - \theta_{\text{joint}}(\lambda), \arctan(d_{\text{stab}}/H_M)\big)$ *(Figure 7(c)).*

Proof. Testing each facet $F \subset \partial\mathcal{G}_1(S)$ across the 880 gems for central symmetry ($F = 2\mathbf{c}(F) - F$ with $c_x(F) = c_y(F) = 0$) identifies the 48 gems in (1) and the 17 gems of projected shoelace area $A_{xy} = 5$ in (2)–(3). For square #763, substituting the lifted coordinates of $\{13, 14, 15, 16\}$ and $\{1, 2, 3, 4\}$ into $4x - 3y + 5(z/\lambda) = 30$ and $-4x - 3y - 5(z/\lambda) = 30$ verifies the facet planes, normals $\mathbf{n}_\pm(\lambda) \propto (\pm 4\lambda, -3\lambda, \pm 5)$, surface area $A_{xy}\sqrt{1 + (4^2 + 3^2)\lambda^2/5^2} = 5\sqrt{1+\lambda^2}$, and joint tilt $\tan\theta_{\text{joint}}(\lambda) = \sqrt{(4\lambda)^2 + (-3\lambda)^2}/5 = \lambda$. □

Remark 5.5 (Three-dimensional chirality and why a single mold cannot build the #763 tower). Because square #763 satisfies $\overline{S} = hS$ ($z(-x, y) = -z(x, y)$), its 3D gem $\mathcal{G}_\lambda(S_{\#763})$ is invariant under the 180° proper rotation around the y-axis, $R_{y,\pi}(x, y, z) = (-x, y, -z) \in SO(3)$, which exchanges $F_+ : 4x - 3y + 5z = 30$ and $F_- : -4x - 3y - 5z = 30$ and returns the *exact same block* in its original orientation. Why, then, can a maker not cast two identical copies of $\mathcal{G}_\lambda(S_{\#763})$ from a single mold and flip the second block upside-down around the x-axis via $R_{x,\pi}(x, y, z) = (x, -y, -z)$ to mate with F_+? Because $R_{x,\pi}$ maps F_+ to a lower facet with outward normal $(+4\lambda, +3\lambda, -5) \neq -\mathbf{n}_+ = (-4\lambda, +3\lambda, -5)$ (reversing the y-slope while keeping the x-slope at $+4\lambda$) and maps F_- to an upper facet with outward normal $(-4\lambda, +3\lambda, +5)$. More fundamentally, the full orthogonal symmetry group of $\mathcal{G}_\lambda(S_{\#763})$ is $\text{Aut}_{O(3)}\big(\mathcal{G}_\lambda(S_{\#763})\big) = \{I_3, R_{y,\pi}\} \cong C_2 \subset SO(3)$, which contains no improper reflections: $\mathcal{G}_\lambda(S_{\#763})$ and its mirror twin $\mathcal{G}_\lambda(S^*_{\#763}) = -\mathcal{G}_\lambda(S_{\#763})$ are *chiral 3D enantiomorphs* lying in disjoint proper rigid-motion orbits ($R\mathcal{G}_\lambda(S_{\#763}) \neq \mathcal{G}_\lambda(S^*_{\#763})$ for all $R \in SO(3)$). Building the alternating tower of Figure 7(a) therefore genuinely requires two opposite-handed molds.

Table 3: Representative order-four Magic Gems from the 17 classes (#603, #763, #768, #774, #779, #785, #788, #791, #792, #818, #824, #828, #836, #837, #839, #842, #844) achieving the maximum projected contact area $A_{xy} = 5$ via centred knight's-move square facets F_+ and F_- (Proposition 5.4). Plane equations (a, b, c, d) are given in doubled coordinates $aX + bY + cZ + d = 0$. "Same-gem ∥" indicates whether $\mathbf{n}_- = -\mathbf{n}_+$; stability margins d_{stab} and single-block critical tipping angles $\theta^{(1)}_{\text{tip}}$ (Definition 5.1) are reported for the upper facet at $\lambda = 0.20$.

1-idx	0-idx	Group	Upper F_+ plane	Lower F_- plane	Same-gem ∥	d_{stab} ($\lambda = 0.2$)	$\theta^{(1)}_{\text{tip}}$ ($\lambda = 0.2$)
#603	602	VI	$(2, -9, 5, -50)$	$(-2, -9, -5, -50)$	No	1.020	45.39°
#763	762	VI	$(4, -3, 5, -60)$	$(-4, -3, -5, -60)$	No	0.969	39.35°
#768	767	II	$(0, -1, 1, -12)$	$(-4, -3, -5, -60)$	No	0.973	39.24°
#774	773	VI	$(1, 0, 1, -12)$	$(-1, 0, -1, -12)$	Yes	0.931	38.35°
#791	790	VI	$(-3, -4, 5, -60)$	$(-3, 4, -5, -60)$	No	0.969	39.35°
#792	791	VI	$(0, -1, 1, -12)$	$(0, 1, -1, -12)$	Yes	0.931	38.35°
#837	836	VI	$(4, -3, 5, -60)$	$(4, 3, -5, -60)$	No	0.962	38.94°

6 Synthesis of Mechanical Censuses and Open Problems

Taken together with the polyhedral classification of [13] and the moment hierarchy of [14], the results of Sections 2–5 show that Magic Gems obey exact mechanical laws across both discrete origin-vector and continuous polyhedral models. Table 4 synthesizes the complete order-four censuses across all four physical regimes, contrasting the universal algebraic invariants shared by all 880 magic squares with the sharp symmetry and Diophantine thresholds that single out specific Dudeney groups and canonical exemplars.

Table 4: Summary of order-four mechanical censuses across the 880 magic squares (certified in exact rational arithmetic; see Section A).

Mechanical Property / Regime	$n = 4$ Count	Exact Extremal / Universal Law	Canonical Exemplars
3D & 2D linear vectors ($\beta = 1, \alpha = 1$)	880/880 (100%)	$\sum \mathbf{p}_{ij}(\lambda) = \mathbf{0}$, $\mathbf{B}_{z,r} = \mathbf{0}$ identically	All semi-magic
3D cubic vector law ($\beta = 3$, $L_{ij} = R^3_{ij}$)	880/880 (100%)	$(\mathbf{V}_3)_z = 0$; $\mathbf{V}_3 = \mathbf{0} \iff$ assoc. (48)	All semi-magic ($z = 0$)
3D unit sphere $\sum \hat{\mathbf{p}}_{ij}(\lambda) = \mathbf{0}$	48/880 (5.5%)	$\mathbf{0}\ \forall\lambda > 0 \iff$ assoc.; 107 have $\beta^* \neq 1$	#175 (**0**), #70 ($\beta^* \approx 2.24$)
3D unit-sphere weights $\mathbf{w}^* > \mathbf{0}$	880/880 (100%)	$\sum w^*_{ij} \hat{\mathbf{p}}_{ij} = \mathbf{0}$, $\|\mathbf{w}^* - \mathbf{1}\|_2 \in [0.005, 1.60]$	All 880 squares
Planar unit-vector norm $\|\mathbf{B}_{\text{unit}}(S)\|_2$	0/880 vanish	$\min = \frac{18\sqrt{10}}{5} - 8\sqrt{2} \approx 0.07049$ ($v/u = 9/4$)	#70, #76, #121, #188, #452, #459, #490, #574
Collinear dipoles & roots $\alpha^* \neq 1$	408/880 (46.4%)	204 with $\kappa > 0$ (25 at $\alpha^* = -1$, 25 at $+3$)	#45 (-1), #286 ($+3$)
Point-mass rotor $I_\perp(\lambda)/I_{zz}$	880/880 (100%)	$\frac{1+17\lambda^2}{2}$ (1 at $\lambda = \frac{1}{\sqrt{17}}$, 9 at $\lambda = 1$)	All semi-magic
Entry-weighted ($m = s$) rotor	56/880 (6.4%)	$Q_{xy} = 0 \implies I_1 = I_2 = 170$ (48 in 3D)	48 III +4 IX +4 X
Naturally balanced solid $\mathbf{c}_0 = \mathbf{0}$	51/880 (5.8%)	48 associative +3 non-assoc. Group VI	#86, #739, #756
2-density core $(\mathcal{G}_0, \mathcal{G}_1)$	35/511 3D cores	33 denser (8 with $\Delta\rho_1/\rho_0 \leq \frac{40}{19}$) +2 cavity	#673 ($\frac{5}{4}$), #123, #54
Non-negative inner-core balance	528/880 (60.0%)	51 natural +477 with $-\mathbf{c}_0 \in \text{cone}(P^{(1)}_S)$	528 magic gems
Complete K_{16} network (120 rods)	880/880 (= mass)	$\mathbf{c} = \mathbf{0}$, $\Sigma_{\text{rods}} = \frac{29}{45}\Sigma(\lambda)$; len. $\|\mathbf{c}\| = 0.025$ ($\lambda = 0.2$)	All 880 (#175)
Nested wireframes $G(\mathcal{G}_0) \cup G(\mathcal{G}_1)$	206/880 ($\lambda = 1$)	48 assoc. +158 Group VI (164 at $\lambda = 0.20$) via d_1/d_0	#673 ($d_1/d_0 = 0.793$)
Papercraft nets & shells $\partial\mathcal{G}_0$	880/880 nets	$N_{\text{tabs}} = f_0 - 1 \in [7, 15]$; $48 + 80$ VI shells balance	#175 (11 tabs, 13 folds)
Resting facet stability	11,044/14,824	74.5% stable at $\lambda = 0.20$ (43.0% at $\lambda = 1$)	$\max \theta_{\text{tip}} = 47.3°$
Mirror-twin vertical towers	48/880 (5.5%)	Max footprint $A_{xy} = 5$ (17 knight's-move)	#763, #603

We close with four open problems suggested by these mechanical findings:

1. **Higher-order Diophantine ring minima.** At order $n = 5$, the 24 non-central cells of A_5 lie on five concentric rings of radii $r \in \{1, \sqrt{2}, 2, \sqrt{5}, 2\sqrt{2}\}$, involving the independent square-free radicals $\sqrt{2}$ and $\sqrt{5}$. Does $\mathbf{B}_{\text{unit}}(S) = \sum z_{ij}\mathbf{u}_{ij}$ vanish for any magic square of order $n \geq 5$, or what is the asymptotic decay rate of $\min_{S \in \mathcal{M}_n} \|\mathbf{B}_{\text{unit}}(S)\|_2$ as $n \to \infty$ via simultaneous Diophantine approximation?

2. **Onion depth of Siamese and linear magic gems.** By [13, Thm. 6.4], the outer hull $\mathcal{G}_0(S)$ of the Siamese magic square of odd order $n \geq 5$ strips off only 12 vertices, leaving $n^2 - 12$ points in $P^{(1)}_S$ lying on five sheared planar polygons. How does the onion peeling depth $d_{\text{onion}}(S_n)$ of the Siamese square grow with n, and what are the asymptotic f-vectors of its inner convex layers $\mathcal{G}_1(S_n) \supset \mathcal{G}_2(S_n) \supset \ldots$?

3. **Exact solid-isotropic Magic Gems.** While point-weighted Magic Gems become exact spherical tops ($I_{xx} = I_{yy} = I_{zz}$) at $\lambda_{\text{iso}} = (n^2 + 1)^{-1/2}$, the uniform polyhedral solid $\mathcal{G}_\lambda(S)$ has a small transverse splitting $(I_3 - I_2)/I_2 \in [0.00004, 0.0243]$ at $n = 4$. Does there exist

a magic square of some order $n \geq 5$ (or a two-density composite $(\mathcal{G}_0, \mathcal{G}_1)$ at $n = 4$) whose continuous solid inertia tensor is an exact scalar multiple of the identity I_3?

4. **Space-filling and interlocking polyhedral packings.** Beyond vertical mirror-twin columns (Theorem 5.3), can a Magic Gem $\mathcal{G}_\lambda(S)$ and its $D_4 \times \langle \tau \rangle$ images tile a horizontal slab or interlock in three-dimensional space with full facet-to-facet contact?

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A Exact computational methods and certificates

All computations in this paper use Python 3.9 with NumPy 2.0, SciPy 1.13, and Python's exact rational module `fractions.Fraction`. The complete verification script `anc/verify_paper_c.py` and figure generator `anc/make_figures_c.py` run in under 10 s on a single core and write the certified census archive `anc/results_paper_c.json`.

Exact simplex moments for continuous polyhedra. As in [13, App. A], doubling the centred coordinates (1) produces integer vectors $P = 2\mathbf{p} \in \mathbb{Z}^3$, and every facet plane $aX + bY + cZ + d = 0$ of $\mathcal{G}_0(S)$ and of the inner core gem $\mathcal{G}_1(S) = \operatorname{conv} P_S^{(1)}$ is certified in exact integer arithmetic. To compute the exact rational volume $\operatorname{vol}(\mathcal{G})$, solid centroid $\mathbf{c}_{\text{solid}}(\mathcal{G})$, and second-moment covariance matrix $\Sigma_{\text{solid}}(\mathcal{G})$ of any 3-dimensional gem $\mathcal{G}$, we decompose $\mathcal{G}$ into tetrahedra $\Delta_k = \operatorname{conv}\{\mathbf{v}_0, \mathbf{v}_{k,1}, \mathbf{v}_{k,2}, \mathbf{v}_{k,3}\}$ sharing the rational vertex $\mathbf{v}_0 = \frac{1}{2}P_0 \in (\frac{1}{2}\mathbb{Z})^3$. Each tetrahedron has exact rational volume

$$V_k = \frac{1}{6}\left|\det(\mathbf{v}_{k,1} - \mathbf{v}_0,\ \mathbf{v}_{k,2} - \mathbf{v}_0,\ \mathbf{v}_{k,3} - \mathbf{v}_0)\right| \in \mathbb{Q}_{\geq 0},$$

exact first moment

$$\iiint_{\Delta_k} \mathbf{x}\, dV = \frac{V_k}{4}\left(\mathbf{v}_0 + \mathbf{v}_{k,1} + \mathbf{v}_{k,2} + \mathbf{v}_{k,3}\right) \in \mathbb{Q}^3,$$

and exact canonical second moment

$$\iiint_{\Delta_k} x_i x_j\, dV = \frac{V_k}{20}\left(2\sum_{a=0}^{3} v_{a,i}v_{a,j} + \sum_{0\leq a\neq b\leq 3} v_{a,i}v_{b,j}\right) \in \mathbb{Q},$$

where $(\mathbf{v}_{k,0}, \mathbf{v}_{k,1}, \mathbf{v}_{k,2}, \mathbf{v}_{k,3}) = (\mathbf{v}_0, \mathbf{v}_{k,1}, \mathbf{v}_{k,2}, \mathbf{v}_{k,3})$. Summing over all simplices Δ_k yields $\operatorname{vol}(\mathcal{G}) = \sum_k V_k \in \mathbb{Q}$, $\mathbf{c}_{\text{solid}}(\mathcal{G}) = \frac{1}{\operatorname{vol}(\mathcal{G})}\sum_k \iiint_{\Delta_k} \mathbf{x}\, dV \in \mathbb{Q}^3$, and $\Sigma_{\text{solid},ij} = \frac{1}{\operatorname{vol}(\mathcal{G})}\sum_k \iiint_{\Delta_k} x_i x_j\, dV - c_{\text{solid},i}c_{\text{solid},j} \in \mathbb{Q}$ without floating-point rounding. Because the simplicial decomposition partitions the convex body $\mathcal{G}$, these rational integrals are independent of the choice of boundary triangulation.

Complete list of the 33 two-density balanced gems. For completeness, Table 5 lists all 33 order-four magic squares for which $\dim \mathcal{G}_1(S) = 3$ and the solid centroids $\mathbf{c}_0(S)$ and $\mathbf{c}_1(S)$ are strictly anti-parallel, together with their exact rational balancing density increment $\Delta\rho_1/\rho_0 = -\operatorname{vol}(\mathcal{G}_0)c_0/(\operatorname{vol}(\mathcal{G}_1)c_1) > 0$ (Theorem 4.3(1)). Every one of the 33 squares belongs to Dudeney's Group VI.

Table 5: Complete census of the 33 order-four Magic Gems admitting exact two-density "diamond-in-a-diamond" solid balance $\mathbf{c}_{\text{comp}} = \mathbf{0}$ (Theorem 4.3(1)), sorted by increasing excess density ratio $\Delta\rho_1/\rho_0$.

Gem	$\text{vol}(\mathcal{G}_0)$	$\text{vol}(\mathcal{G}_1)$	$\Delta\rho_1/\rho_0$	Gem	$\text{vol}(\mathcal{G}_0)$	$\text{vol}(\mathcal{G}_1)$	$\Delta\rho_1/\rho_0$	Gem	$\text{vol}(\mathcal{G}_0)$	$\text{vol}(\mathcal{G}_1)$	$\Delta\rho_1/\rho_0$
#673	257/3	56/3	5/4	#665	281/3	6	61/13	#697	248/3	28/3	166/11
#502	271/3	6	4/3	#252	233/3	7	88/9	#671	84	2/3	31/2
#45	214/3	6	85/61	#166	93	8/3	21/2	#383	284/3	7	47/3
#50	254/3	22/3	28/19	#158	88	9	57/5	#398	260/3	7	95/6
#307	277/3	22/3	23/15	#168	280/3	7	110/9	#18	71	9	33/2
#49	81	2	2	#672	260/3	4/3	99/8	#627	83	4/3	67/4
#38	263/3	22/3	31/15	#167	90	8/3	27/2	#452	236/3	6	118/7
#47	247/3	22/3	40/19	#199	256/3	22/3	206/15	#604	221/3	26/3	229/10
#261	259/3	7	47/12	#615	244/3	5	14	#226	235/3	7	143/6
#670	244/3	10	103/25	#6	239/3	7	127/9	#626	69	2/3	49/2
#488	277/3	22/3	71/16	#573	251/3	8	44/3	#436	235/3	6	191/6